Renormalization of 3D Luttinger action

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Three-dimensional Luttinger fermions have quadratic band touching dispersion with Lifshitz-like dynamical scaling $z = 2$. We extend the single-particle Luttinger Hamiltonian to include the effects of minimally coupled gauge fluctuations and illustrate the long-wavelength behavior of the combined interacting system. Within the parameter space of our model, we discover an analog of the non-Fermi liquid phase found in Luttinger semimetals with static Coulomb interaction [1]. We further show that dynamical gauge fluctuations can also lead to non-relativistic Lifshitz scaling for the emergent photons in our theory. With an RG analysis, we describe the stability and scaling properties of its fixed points in the presence of additional cubic-symmetry allowed perturbations.

I. THE ACTION AND ITS CONSERVED CURRENTS

A. The fermion sector

Let us consider a U(1) gauge theory coupled to fermions with symmetry protected quadratic band touching. The action without the gauge fields has been discussed before [1, 2]. In the rotation and particle-hole symmetric limit, using $d_\gamma = (d-1)(d+2)/2$ Clifford matrices $\{\gamma^a\}$ of dimensions $2^{\lfloor d_\gamma/2 \rfloor} \times 2^{\lfloor d_\gamma/2 \rfloor}$, the Euclidean Lagrangian is expressed as [3, 4]:

$$L_\psi^{(0)} = \psi^\dagger \left[\partial_0 - \frac{1}{2m} \sum_{a=1}^{d_\gamma} \gamma^a g_a(\partial, \partial) \right] \psi,$$
$$g_a(\partial, \partial) = \sqrt{\frac{d}{2(d-1)}} \sum_{i,j=1}^{d} \lambda_a^{ij} \frac{\partial^2}{\partial x_i \partial x_j}, \ \{\gamma^a, \gamma^b\} = 2\delta_{ab}, \tag{1}$$

where $\{\lambda_a\}$ are the d_γ real, symmetric $d \times d$ Gell-Mann matrices [3, 4] satisfying the anticommutation relationships,

$$\{\lambda_b, \lambda_c\}^{ij} = \frac{4}{d} \delta_{bc} \delta^{ij} + 2 d_{abc} \lambda_a^{ij}. \tag{2}$$

We introduce the Clifford algebra valued metric tensor, which we write with the Dirac slash $P_a \gamma^a = \not{P}$,

$$\not{g}^{ij} = \sum_{a=1}^{d_\gamma} g_a^{ij} \gamma^a, \quad g_a^{ij} = \sqrt{\frac{d}{2(d-1)}} \, \lambda_a^{ij}. \tag{3}$$

We keep the spatial dimension d as a free parameter, and the spinor trace thus carries the d-dependent factor $2^{\lfloor d_\gamma/2 \rfloor}$ until the expansion in $d = 4 - \epsilon$ is performed at the very last stage.[1] In

[1] In three dimensional momentum space the band dispersions are [4] $g_1(k) = \frac{\sqrt{3}}{2}\left(k_x^2 - k_y^2\right)$, $g_2(k) = \sqrt{3} k_x k_y$, $g_3(k) = \sqrt{3} k_x k_z$, $g_4(k) = \sqrt{3} k_y k_z$, and $g_5(k) = \frac{1}{2}\left(2k_z^2 - k_x^2 - k_y^2\right)$.

our notation the greek indices span the space-time and the roman indices only span the space. It can be shown that $\gamma^{ab} = \frac{1}{2i}[\gamma^a, \gamma^b]$ are the $(d-1)(d+2)\left(d^2+d-4\right)/8$ generators of the rotations among them [5]. Certain useful identities of the γ matrices are as follows:

$$\begin{aligned}
\sum_{b=1}^{d_\gamma} \gamma^b \gamma^a \gamma^b &= -\frac{(d+3)(d-2)}{2}\gamma^a \\
\sum_{c=1}^{d_\gamma} \gamma^c \gamma^a \gamma^b \gamma^c &= \frac{d^2+d-10}{2}\gamma^a\gamma^b + 4\delta_{ab} \\
\sum_{e=1}^{d_\gamma} \gamma^e \gamma^a \gamma^b \gamma^c \gamma^e &= -2\gamma^c\gamma^b\gamma^a - \frac{d^2+d-10}{2}\gamma^a\gamma^b\gamma^c.
\end{aligned} \tag{4}$$

The symmetric Gell-Mann matrices obey the identities [2–4],

$$\begin{aligned}
&\sum_{a=1}^{d_\gamma} \lambda_a^{ij}\lambda_a^{kl} = (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) - \frac{2}{d}\delta_{ij}\delta_{kl}, \\
&\mathrm{tr}\left[\lambda_a\lambda_b\right] = 2\delta_{ab}.
\end{aligned} \tag{5}$$

Using these identities it can be shown in a simple way that [4]

$$\begin{aligned}
&\sum_{a=1}^{d_\gamma} g_a(p)g_a(q) = \frac{d}{d-1}\left((\vec{p}\cdot\vec{q})^2 - \frac{1}{d}p^2q^2\right), \\
\implies &\sum_{a=1}^{d_\gamma} g_a(q)g_a(q) = q^4.
\end{aligned} \tag{6}$$

The global U(1) symmetry of the action (1) is manifest with the global transformations $\psi \mapsto e^{ie\alpha}\psi$. The corresponding global currents are given by,

$$\begin{aligned}
j_0 &= i\psi^\dagger\psi, \\
j_i &= i\sqrt{\frac{d}{2(d-1)}}\sum_{a=1}^{d_\gamma}\sum_{j=1}^{d}\frac{\lambda_a^{ij}}{2m}\left[\psi^\dagger\gamma^a\frac{\partial}{\partial x_j}\psi - \frac{\partial}{\partial x_j}\psi^\dagger\gamma^a\psi\right],
\end{aligned} \tag{7}$$

with the current conservation condition,

$$\sum_{\mu=0}^{d}\partial_\mu j_\mu = 0, \tag{8}$$

dictated by the symmetry. We seek to elevate the global U(1) symmetry of the model to a local U(1) gauge symmetry by introducing vector gauge fields A_μ which are minimally coupled with the action. The inclusion of the minimal coupling is achieved by replacing all the normal derivatives of the action with covariant derivatives $D_\mu = \partial_\mu + ieA_\mu$. With the covariant derivatives, the action reads as:

$$L_\psi = \psi^\dagger\left[\partial_0 + ieA_0 - \frac{\not{p}(D,D)}{2m}\right]\psi, \tag{9}$$

with the rule that under the local U(1) transformations both the fermionic fields and the vector gauge fields transform simultaneously with the formula,

$$\begin{aligned} &\mathrm{U}(1) : \psi \mapsto e^{ie\varphi(x)}\psi, \\ &\mathrm{U}(1) : A_\mu(x) \mapsto A_\mu(x) - \partial_\mu\varphi(x), \end{aligned} \tag{10}$$

where $x = (\tau, \vec{x})$ denotes the space-time coordinate. Similarly for the frequency-momentum space we shall use the notation $k = (k_0, \vec{k})$. It is straightforward to check that under the gauge transformations [Eq. (10)] the action of the covariant derivative changes as,

$$\mathrm{U}(1) : D_\mu\psi \mapsto (\partial_\mu + ieA_\mu(x) - ie\partial_\mu\varphi(x))e^{ie\varphi(x)}\psi = e^{ie\varphi(x)}D_\mu\psi, \tag{11}$$

which in turn ensures that the minimally coupled action written down above (9) remains invariant under the gauge transformation.

B. The photon sector

With the fermionic sector well defined we need a gauge invariant description of the gauge field action. We are interested in a U(1) gauge theory in (3+1) dimensions. The gauge invariant Euclidean Maxwell action is given by [5, 6]

$$L_{\mathrm{Maxwell}} = \frac{1}{4}F_{\mu\nu}^2 + ej^\mu A_\mu, \tag{12}$$

with the matter current j^μ defined above (7). However, we are in a situation where the Lorentz invariance is absent in the fermionic sector, and naturally, space and time have different engineering dimensions. This scenario has been explored in the context of the pure U(1) gauge theory of pyrochlore compounds [7, 8]. From the fermion sector it is imperative that before any loop correction the engineering dimensions of space and time are respectively $[x^{-1}] = 1$ and $[\tau^{-1}] = z_0 = 2$. The appropriate U(1) gauge theory in this situation goes by the name of $z_0 = 2$ Lifshitz electrodynamics [9, 10] and its properties have been studied in detail [9–12]. The Lagrangian is most compactly written in Euclidean space-time as,

$$L_{\mathrm{U}(1)} = \frac{1}{2}F_{0i}F_{0i} + \frac{1}{4}F_{ij}\left(c^2 + v^2\left[-\nabla^2\right]\right)F_{ij} + ej_\mu A_\mu, \text{ with } F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{13}$$

As a consequence of gauge invariance, the photon propagator obtained from this action is singular. To determine the photon propagator we need to work in a fixed gauge. In a Lorentz invariant theory a gauge fixing function $G(A) = \partial_\mu A_\mu$ can be set to zero because under a gauge transformation $A_\mu(x) \mapsto A_\mu^\alpha(x) = A_\mu(x) + \partial_\mu\alpha(x)$ the function transforms as

$$G(A_\mu^\alpha(x)) = \partial_\mu A_\mu(x) + \partial^2\alpha(x), \tag{14}$$

and we can always choose an $\alpha(x)$ such that the function vanishes. The gauge fixing function can in principle be completely arbitrary. In the current case a suitable gauge fixing function for the action is given by [10],

$$G(A_\mu, \omega) = \partial_0 A_0(x) + \Delta_{\mathrm{g}}\frac{\partial}{\partial x_i}A_i(x) - \omega(x) = 0, \tag{15}$$

where $\omega(x)$ is an arbitrary function and we have defined a differential operator $\Delta_{\rm g} = c^2\Lambda^2 + v^2\left[-\nabla^2\right]$. In the path integral we must only sum over the unique gauge configurations dictated by the gauge orbit $\omega(x)$ (15). Various different gauge field configurations which are related to each other through the $\alpha(x)$ reparameterization must be counted once. The original functional integral is given by

$$Z_{\rm U(1)}[j] = \int \mathcal{D}A e^{-\int_x L_{\rm U(1)}(A_\mu, j_\mu)}, \tag{16}$$

where $\int_x = \int dx_0 d^d x$ is a short-hand notation for the $d+1$ dimensional space-time integral. We wish to integrate over gauge field configurations which satisfy the gauge fixing condition $G(A_\mu, \omega) = 0$. We shall therefore sum over all gauge field reparameterizations $A_\mu(x) \mapsto A^\alpha_\mu(x)$ that leave the functional integral invariant and also obey the gauge fixing condition. This can be carried out by the Faddeev-Popov prescription where one uses the identity [13],

$$1 = \int \mathcal{D}\alpha \det\left(\frac{\delta G(A^\alpha_\mu, \omega)}{\delta\alpha}\right) \delta\left(G(A^\alpha_\mu, \omega)\right), \tag{17}$$

to modify the partition function of the gauge fields with a uniform unity weight,

$$Z_{\rm U(1)}[j] = \int \mathcal{D}\alpha\mathcal{D}A e^{-\int_x L_{\rm U(1)}(A_\mu, j_\mu)} \det\left(\frac{\delta G(A^\alpha_\mu, \omega)}{\delta\alpha}\right) \delta\left(G(A^\alpha_\mu, \omega)\right). \tag{18}$$

With

$$\frac{\delta G(A^\alpha_\mu, \omega)}{\delta\alpha} = \partial_0^2 + \Delta_{\rm g}\frac{\partial^2}{\partial x_i^2} \tag{19}$$

the determinant in this expression factors out. With the intention to make the photon propagator simpler, a non-local gauge orbit can be chosen for our Lorentz-violating gauge field action [10–12] with a gauge fixing function,

$$G(A_\mu, \varphi) = \partial_0 A_0(x) + \Delta_{\rm g}\frac{\partial}{\partial x_i}A_i(x) - \Delta_{\rm g}^{1/2}\varphi(x), \tag{20}$$

where we shall sum over all functions $\varphi(x)$ with a Gaussian weight centered around $\varphi(x) = 0$. Finally, we make a gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu\alpha$ in the functional integral over the gauge fields to obtain,

$$\begin{aligned} Z_{\rm U(1)}[j] =& \int \mathcal{D}\varphi e^{-\int_x \frac{\varphi(x)^2}{2\xi}} \int \mathcal{D}\alpha \det\left(\partial_0^2 + \Delta_{\rm g}\frac{\partial^2}{\partial x_i^2}\right) \\ & \times \int \mathcal{D}A^\alpha e^{-\int_x L_{\rm U(1)}(A^\alpha_\mu, j_\mu)} \delta\left(\partial_0 A^\alpha_0(x) + \Delta_{\rm g}\frac{\partial}{\partial x_i}A^\alpha_i(x) - \Delta_{\rm g}^{1/2}\varphi(x)\right) \\ =& \det\left(\partial_0^2 + \Delta_{\rm g}\frac{\partial^2}{\partial x_i^2}\right) \int \mathcal{D}\alpha \int \mathcal{D}A^\alpha e^{-\int_x L_{\rm U(1)}(A^\alpha_\mu, j_\mu)} e^{-\int_x \frac{\left(\Delta_{\rm g}^{-1/2}\partial_0 A^\alpha_0(x) + \Delta_{\rm g}^{1/2}\frac{\partial}{\partial x_i}A^\alpha_i(x)\right)^2}{2\xi}}. \end{aligned} \tag{21}$$

It is clear in the second line that $\alpha(x)$ has become a dummy parameter and naturally the functional integral over $\alpha(x)$ simply factors out. The procedure therefore achieves its goal of summing over

unique gauge field configurations. The Feynman-'t Hooft gauge $\xi = 1$ further simplifies the gauge fixed action,

$$L_{\rm gf} = \frac{1}{2}F_{0i}F_{0i} + \frac{1}{4}F_{ij}\Delta_{\rm g}F_{ij} + \frac{1}{2}\left(\Delta_{\rm g}^{-1/2}\partial_0 A_0 + \Delta_{\rm g}^{1/2}\frac{\partial}{\partial x_i}A_i\right)^2, \tag{22}$$

and yields a simple non-singular gauge field propagator [11],

$$D^{\rm gf}_{\mu\nu}(k) = \langle A_\mu(-k)A_\nu(k)\rangle = \left(\delta_{\mu 0}\delta_{\nu 0}\frac{\Delta_{\rm g}(k^2)}{k_0^2 + k^2\Delta_{\rm g}(k^2)} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{k_0^2 + k^2\Delta_{\rm g}(k^2)}\right), \tag{23}$$

where $k^2 = k_i^2$ is the squared length of the momentum $\vec{k}$. Henceforth, we shall use the gauge-fixed propagator $D^{\rm gf}$ without referring to the gauge fixing directly.

II. RG ANALYSIS

A. The action

The renormalization group analysis of the fully interacting problem begins by separating the action into the free part and the interacting vertices. For this analysis we must consider a gauge-invariant Lagrangian with symmetry allowed terms that are closed under renormalization. For this reason, in addition to the terms discussed so far we also include the Pauli, or Zeeman, coupling term,

$$-\frac{g}{2}\psi^\dagger\Sigma^{ij}\psi F_{ij}, \tag{24}$$

with

$$\Sigma^{ij} = -\Sigma^{ji} = \frac{1}{4}\left[\lambda_a, \lambda_b\right]^{ij}\gamma^{ab}, \quad \gamma^{ab} = \frac{1}{2i}\left[\gamma^a, \gamma^b\right], \tag{25}$$

the generator of $SO(d)$ spatial rotations in the $2^{\lfloor d_\gamma/2\rfloor}$ dimensional spinor representation.[2] At $d = 3$ one has $\Sigma^{ij} = \epsilon_{ijk}S^k$ with S^i the spin 3/2 matrices, so that Eq. (24) is the familiar $-g\,\psi^\dagger\vec{S}\psi\cdot\vec{B}$.[3] Here $\eta^i_{ab} = -\eta^i_{ba}$ are the symbols that relate Spin 3/2 SU(2) matrices with γ_{ab} [2],

$$S^i = \eta^i_{ab}\gamma^{ab}, \left[\eta^i, \eta^j\right] = -\frac{1}{4}\epsilon^{ij}_k\eta^k.$$

We shall need some more properties of η^i matrices from their defining relation. For $D = 4$ dimensional γ^{ab} matrices we have the formula

$$\mathrm{tr}\left[\gamma^{ab}\gamma^{cd}\right] = D\left(\delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc}\right), \tag{26}$$

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[2] We use Eq. (25) for general d and fix it by requiring that Σ^{ij} rotate ϕ^{ij} as a rank-two tensor, as we work out in the companion Zeeman note.

[3] At $d = 3$ alone we may write the same coupling as $+\frac{\kappa}{2}\psi^\dagger\gamma^{ab}\psi\,\eta^i_{ab}\epsilon_{ijk}F_{jk}$ with $\kappa = -g$.

which we may use to find,

$$\begin{aligned}\mathrm{tr}\left[S^i S^j\right] &= -2D\eta^i_{ab}\eta^j_{ba},\\ \mathrm{tr}\left[\eta^i\eta^j\right] &= -\frac{5}{2D}\delta^{ij} = -\frac{5}{8}\delta^{ij}.\end{aligned} \tag{27}$$

It also follows that $\eta^i\eta^i = cI$, because this operator commutes with all the generators η^i. Using the trace identity we have,

$$\eta^i\eta^i = -\frac{3}{8}I. \tag{28}$$

To relate η^i_{ab} symbols with the λ^{ij}_a matrices we need the formula [14],

$$\gamma^a = \frac{1}{\sqrt{2d(d-2)}}\{S^i, S^j\}\lambda^{ij}_a, \tag{29}$$

and use it in conjunction with the η and λ matrices. We find,

$$\begin{aligned}\eta^i_{ab} &= \frac{1}{8}\mathrm{Tr}\left[S^i\gamma^{ab}\right],\\ &= \frac{1}{32d(d-2)i}\lambda^{mn}_a\lambda^{rs}_b\mathrm{Tr}\left[S^i\left[\{S^m,S^n\},\{S^r,S^s\}\right]\right].\end{aligned} \tag{30}$$

The S^i commutators in the last line can be extracted by inverting Eq. (29) above. The Zeeman coupling g has dimension $[g] = (3z_0 - d - 2)/2$ and is relevant in the infrared for $d < 4$. Together, we have the action,

$$\begin{aligned}L =& L_{B,A} + L^{(0)}_{B,\psi} + L_{B,\psi A},\\ L_{B,A} =& \frac{1}{2}F_{B,0i}F_{B,0i} + \frac{1}{4}F_{B,ij}\left(c_B^2 + v_B^2\left[-\nabla^2\right]\right)F_{B,ij},\\ L^{(0)}_{B,\psi} =& \psi^\dagger_B\left[\partial_0 - \frac{\not{g}(\partial,\partial)}{2m_B}\right]\psi_B,\\ L_{B,\psi A} =& ie_B\left[\psi^\dagger_B A_{B,0}\psi_B - \frac{1}{2m_B}\psi^\dagger_B\not{g}(A_B, \overleftrightarrow{\partial})\psi_B\right]\\ &+ \frac{e_B^2}{2m_B}\psi^\dagger_B\not{g}\left(A_B, A_B\right)\psi_B - \frac{g_B}{2}\psi^\dagger_B\Sigma^{ij}\psi_B F_{B,ij},\end{aligned} \tag{31}$$

where $c_B^2 > 0, v_B^2 > 0$ for a positive definite (bounded) bosonic action, and we have introduced the Dirac slash notation, $P_a\gamma^a = \not{P}$. The two-sided derivative is

$$\psi^\dagger\overleftrightarrow{\partial}_j\psi = \psi^\dagger(\partial_j\psi) - (\partial_j\psi^\dagger)\psi, \tag{32}$$

and with the form

$$\not{g}(\mathrm{d}\boldsymbol{\xi}_1, \mathrm{d}\boldsymbol{\xi}_2) = \sum_{i,j}\not{g}^{ij}\xi_{1,i}\xi_{2,j} \tag{33}$$

of Eq. (3) and $D_\mu = \partial_\mu + ie_B A_{B,\mu}$ the fermion terms other than the Σ^{ij} term are, up to a total derivative,

$$\psi_B^\dagger \left[D_0 - \frac{\not{g}(D,D)}{2m_B} \right] \psi_B. \tag{34}$$

The zero temperature Euclidean theory is embedded in $(d+1)$ dimensions. The tree level engineering dimensions of space and time can be set at $[x] = -1$ and $[\tau] = -z_0$. The engineering dimensions of the various fields and couplings of the theory are as follows,

$$\begin{aligned} &[\psi] = d/2,\ \left[m^{-1}\right] = z_0 - 2,\ [A_0] = (d + z_0 - 2)/2,\ [A_i] = (d - z_0)/2, \\ &\left[e^2\right] = 2 + z_0 - d,\ \left[c^2\right] = 2z_0 - 2,\ \left[v^2\right] = 2z_0 - 4, \\ &[g] = (3z_0 - d - 2)/2. \end{aligned} \tag{35}$$

With $z_0 = 2$ all the couplings of the action are dimensionless except for e^2 and c.[**ToDo: Hunch: at $z_0 = 2$, $[g] = (4-d)/2 \neq 0$ for $d \neq 4$; cf. the text after Eq. (29).**] It is clear that the theory is strongly coupled with the relevant coupling c in any spatial dimension with $z_0 = 2$. On the other hand the choice of $z_0 = 1$ turns the fermion bands flat. To investigate whether the theory has a Wilsonian fixed point within its parameter space we manually tune the relevant coupling to a critical value $c_c = 0$ and study the effective theory within a critical hypersurface parameterized by the remaining marginal couplings. The critical gauge field action therefore becomes,

$$L_A^c = \frac{1}{2} F_{B,0i} F_{B,0i} + \frac{v_B^2}{4} F_{B,ij} \left[-\nabla^2\right] F_{B,ij}. \tag{36}$$

However, the gauge charge e is also strongly relevant in the physical spatial dimension of interest $d = 3$. For perturbative control on the effective critical theory we follow Moon et al. [1] to work in $d = 4 - \epsilon$ dimensions such that all the couplings of the theory are marginally relevant or irrelevant except for c which we critically tune to zero. Along with the compact temporal integral (Euclidean time-cycle) we are therefore undertaking a $5 - \epsilon$ expansion of the theory. We start by renormalizing the bare fields and parameters as,

$$\begin{aligned} &\psi_B = \sqrt{Z_\psi}\psi,\ A_{B,\mu} = \sqrt{Z_A} A_\mu, \\ &m_B^{-1} = Z_{m^{-1}} m^{-1},\ e_B \mapsto Z_e e,\ v_B \mapsto Z_v v,\ c_B = Z_c c \Lambda, \\ &g_B = Z_g g. \end{aligned} \tag{37}$$

[**ToDo: Hunch: $e_B = Z_e e$ and $v_B = Z_v v$.**] The renormalized Lagrangian, in which we keep the Amperean term apart from the critical action as the inserted operator O_c, is therefore given by,

$$\begin{aligned} L_{\text{crit}} &= L_A^c + L_\psi^{(0)} + L_{\psi A} + O_c, \\ L_A^c &= \frac{Z_A}{2} F_{0i} F_{0i} + \frac{Z_A Z_v^2 v^2}{4} F_{ij} \left[-\nabla^2\right] F_{ij}, \\ O_c &= \frac{Z_A Z_c^2 c^2 \Lambda^2}{4} F_{ij} F_{ij}, \\ L_\psi^{(0)} &= Z_\psi \psi^\dagger \left[\partial_0 - Z_{m^{-1}} \frac{\not{g}(\partial,\partial)}{2m} \right] \psi, \\ L_{\psi A} &= Z_e Z_\psi \sqrt{Z_A}\, ie \left[\psi^\dagger A_0 \psi - \frac{Z_{m^{-1}}}{2m} \psi^\dagger \not{g}(A, \overleftrightarrow{\partial}) \psi \right] \end{aligned}$$

$$+ Z_e^2 Z_\psi Z_A Z_{m^{-1}} \frac{e^2}{2m} \psi^\dagger \not{g}(A, A)\, \psi - Z_g Z_\psi \sqrt{Z_A} \frac{g}{2} \psi^\dagger \Sigma^{ij} \psi F_{ij}. \tag{38}$$

The coefficient $c^2\Lambda^2$ of O_c is dimensionful, a Wilsonian mass of the photon. We keep it out of the critical action and admit it into the perturbation series only through insertions: we take the photon propagator below from Eq. (23) with $\Delta_{\rm g}(k^2) = v^2k^2$, which we write as ρk^2 after the rescaling of Eq. (43). Since $\Delta_{\rm g}$ contains $c^2\Lambda^2$, we also find it in the gauge fixing term of Eq. (22), and we insert the whole c-dependent part of $L_{\rm gf}$,

$$L_{\rm gf} - L_{\rm gf}|_{c=0} = \frac{c^2\Lambda^2}{2} \left\{ A_0 \left(\frac{\partial_0^2}{v^4\nabla^4} \right) A_0 + A_i \left(-\nabla^2 \right) A_i \right\} + O(c^4\Lambda^4), \tag{39}$$

whose $F_{ij}F_{ij}$ part is O_c at tree level; we take the remainder from the gauge fixing term, which we do not renormalize. We treat the fermion bilinears $s\, i\psi^\dagger\psi$ and $t_a \psi^\dagger \gamma^a \psi$ of Sec. V E as inserted operators in the same sense. Gauge invariance imposes further constraints on the renormalization parameters. We have,

$$\begin{aligned} & Z_\psi = Z_e Z_\psi \sqrt{Z_A} \text{ and } Z_\psi Z_{m^{-1}} = Z_e^2 Z_\psi Z_A Z_{m^{-1}}, \\ \Longrightarrow\ & Z_e \sqrt{Z_A} = 1. \end{aligned} \tag{40}$$

We redefine the remaining independent renormalization parameters as,

$$\begin{aligned} Z_\psi &= Z_1, \\ Z_\psi Z_{m^{-1}} &= Z_2, \\ Z_A &= Z_3, \\ Z_A Z_c^2 &= Z_4, \\ Z_A Z_v^2 &= Z_5, \\ Z_g Z_\psi \sqrt{Z_A} &= Z_6. \end{aligned} \tag{41}$$

Of these, we use Z_4 to renormalize the inserted operator O_c and not a term of the critical action. As a consequence it is possible to find the one-loop renormalization group coefficients of the theory by studying its two-point functions alone.

B. Dimensionless couplings

We use one set of dimensionless couplings,

$$\alpha = \frac{me^2}{8\pi^2}, \quad K = m^2c^2, \quad \rho = m^2v^2, \quad \sigma = \frac{g^2m^3}{8\pi^2}, \tag{42}$$

where $c\Lambda$ is the photon velocity, v the photon stiffness and g the Zeeman coupling; we place Λ^2 in front of the $\boldsymbol{B}^2$ term to make c dimensionless. Once we scale the band mass out of the action, we are left with these couplings alone. Let us measure the imaginary time in units of m and rescale the gauge field with it,

$$x_0 = m\, x_0', \quad A_0 = \frac{A_0'}{\sqrt{m}}, \quad A_i = \sqrt{m}\, A_i'. \tag{43}$$

With this change we multiply the one-form $A_\mu \mathrm{d}x_\mu$ by $\sqrt{m}$ and leave ψ unchanged. Dropping the primes, we write the renormalized Lagrangian of Eq. (38) with the renormalization parameters of Eq. (41) as

$$
\begin{aligned}
L &= Z_1 \psi^\dagger D_0 \psi - Z_2 \psi^\dagger \frac{\not{g}(D,D)}{2} \psi + \frac{Z_3}{2} F_{0i} F_{0i} + \frac{Z_5 \rho}{4} F_{ij} \left[-\nabla^2\right] F_{ij} - Z_6 \frac{\sqrt{8\pi^2\sigma}}{2} \psi^\dagger \Sigma^{ij} \psi F_{ij} + O_c, \\
O_c &= \frac{Z_4 K \Lambda^2}{4} F_{ij} F_{ij}, \quad D_\mu = \partial_\mu + i\sqrt{8\pi^2\alpha}\, A_\mu,
\end{aligned} \tag{44}
$$

for $g \geq 0$, and the gauge fixing operator as $\Delta_\mathrm{g} = K\Lambda^2 + \rho\left[-\nabla^2\right]$. From here on we write s and t_a for the inserted couplings ms and mt_a. Furthermore we shall expand around the tree level values of these renormalization factors, $Z_i = 1 + \delta Z_i = 1 + \delta_i$. This splits the Lagrangian between its tree level form and counter terms:

$$
\begin{aligned}
L =& L_A^c + L_\psi^{(0)} + L_{\psi A} + O_c + L_\mathrm{ct}, \\
L_A^c =& \frac{1}{2} F_{0i} F_{0i} + \frac{\rho}{4} F_{ij} \left[-\nabla^2\right] F_{ij}, \\
O_c =& \frac{K\Lambda^2}{4} F_{ij} F_{ij}, \\
L_\psi^{(0)} =& \psi^\dagger \left[\partial_0 - \frac{\not{g}(\partial,\partial)}{2}\right] \psi, \\
L_{\psi A} =& i\sqrt{8\pi^2\alpha}\Big[\psi^\dagger A_0 \psi - \frac{1}{2}\psi^\dagger \not{g}(A, \overleftrightarrow{\partial})\psi\Big] + \frac{8\pi^2\alpha}{2} \psi^\dagger \not{g}(A,A)\psi \\
& - \frac{\sqrt{8\pi^2\sigma}}{2} \psi^\dagger \Sigma^{ij} \psi F_{ij}, \\
L_\mathrm{ct} =& \delta_3 \frac{1}{2} F_{0i} F_{0i} + \delta_4 \frac{K\Lambda^2}{4} F_{ij} F_{ij} + \delta_5 \frac{\rho}{4} F_{ij} \left[-\nabla^2\right] F_{ij} \\
& + \delta_1 \psi^\dagger D_0 \psi - \delta_2 \psi^\dagger \frac{\not{g}(D,D)}{2} \psi - \delta_6 \frac{\sqrt{8\pi^2\sigma}}{2} \psi^\dagger \Sigma^{ij} \psi F_{ij}.
\end{aligned} \tag{45}
$$

We have all the pieces in hand to evaluate the renormalized perturbation series diagrammatically in the momentum space. We adopt the following convention for the Fourier transformation of the fields,

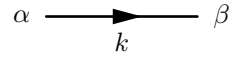
$$
A_\mu(x) = \int \frac{d^{d+1}k}{(2\pi)^{d+1}} A_\mu(k) e^{ik\cdot x}, \tag{46}
$$

$$
\psi(x) = \int \frac{d^{d+1}k}{(2\pi)^{d+1}} \psi(k) e^{ik\cdot x}. \tag{47}
$$

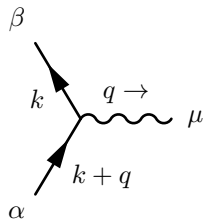
As a consistent choice of notation, momentum entering a Feynman diagram vertex shall be considered positive, while the momentum leaving a vertex shall be considered negative. Defining a function,

$$
\not{g}^i(k) = \not{g}(k, \mathrm{d}x_i) = \sqrt{\frac{d}{2(d-1)}} \gamma^{ij} k_j, \tag{48}
$$

with $\not{g}(k) = \not{g}(k,k)$, we can compactly write down the momentum space Feynman rules for the action:

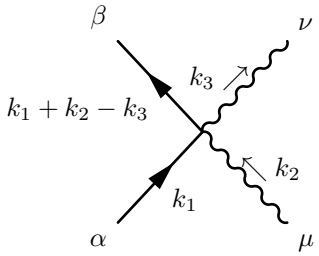
α  β $= \langle \psi_\alpha(k)\psi^\dagger_\beta(k)\rangle = G_{\alpha\beta}(k) = \left[\dfrac{-ik_0 + \not{g}(k)/2}{k_0^2 + k^4/4}\right]_{\alpha\beta}$

μ  ν $= \langle A_\mu(-k)A_\nu(k)\rangle = D^{\rm gf}_{\mu\nu}(k) = \left(\delta_{\mu 0}\delta_{\nu 0}\dfrac{\rho k^2}{k_0^2+\rho k^4} + \delta_{\mu i}\delta_{\nu j}\dfrac{\delta_{ij}}{k_0^2+\rho k^4}\right),$



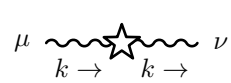
$= C^\mu_{\alpha\beta}(k+q,-q,-k) = -\sqrt{8\pi^2\alpha}\Big[i\delta_{\mu 0} + \dfrac{\not{g}^i(2k+q)}{2}\delta_{\mu i}\Big]_{\alpha\beta}$

$+ \sqrt{8\pi^2\sigma}\left[\Sigma^{jk}\right]_{\alpha\beta}(-iq_j)\,\delta_{\mu k},$

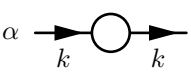


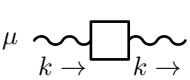
$= V^{\mu\nu}_{\alpha\beta}(k_1,k_2,-k_3,-k_1-k_2+k_3) = -8\pi^2\alpha\sqrt{\dfrac{d}{2(d-1)}}\delta_{\mu i}\delta_{\nu j}\left[\not{\chi}^{ij}\right]_{\alpha\beta},$

α  β $= -\left[is + t_a\gamma^a\right]_{\alpha\beta},$

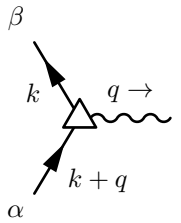
μ  ν $= O^{\mu\nu}_c(k) = K\Lambda^2\left\{\delta_{\mu 0}\delta_{\nu 0}\dfrac{k_0^2}{\rho^2 k^4} - \delta_{\mu i}\delta_{\nu j}\delta_{ij}k^2\right\},$ (49)

where the factor of 2 in the quartic vertex comes from the symmetry permutation of the photon fields. The Feynman diagrams of the counter term vertices are as follows:

α  β $= \delta_\Sigma = -\left[\delta_1\,(ik_0) + \delta_2\left(\dfrac{\not{g}(k)}{2}\right)\right]_{\alpha\beta},$

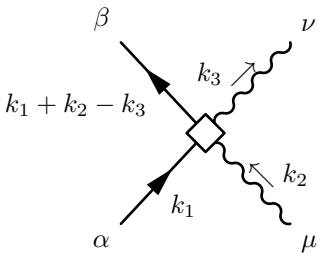
μ  ν $= \delta_\Pi = -\delta_3\left\{\delta_{\mu 0}\delta_{\nu 0}k^2 - \delta_{\mu 0}\delta_{\nu j}k_0k_j - \delta_{\mu i}\delta_{\nu 0}k_ik_0 + \delta_{\mu i}\delta_{\nu j}\delta_{ij}k_0^2\right\}$

$$- \delta_4 \delta_{\mu i} \delta_{\nu j} K \Lambda^2 \left(\delta_{ij} k^2 - k_i k_j \right) - \delta_5 \delta_{\mu i} \delta_{\nu j} \rho k^2 \left(\delta_{ij} k^2 - k_i k_j \right),$$



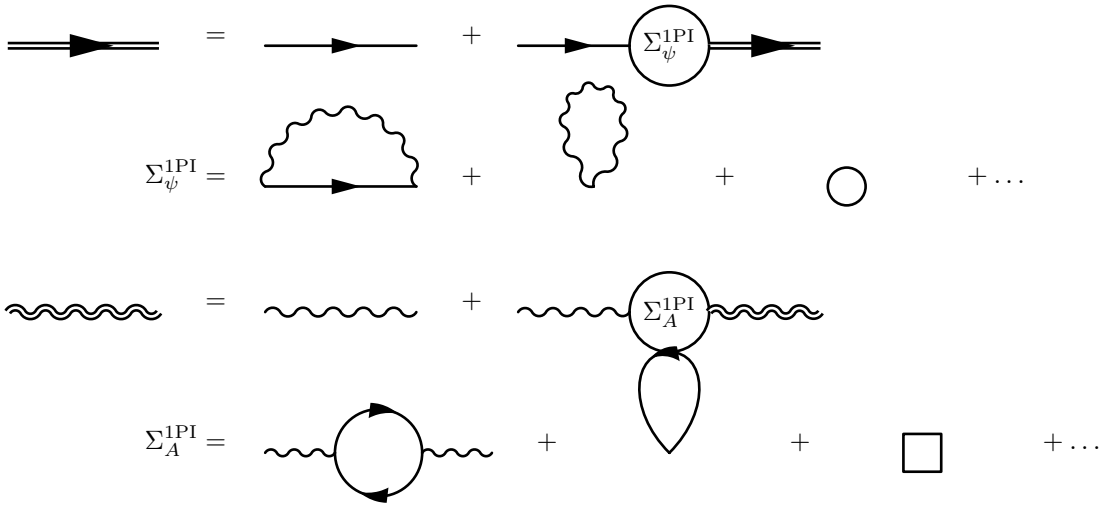
$$= \delta_C = -\sqrt{8\pi^2\alpha}\Big[\delta_1 \left(i\delta_{\mu 0}\right) + \delta_2 \left(\frac{\not{q}^i (2k+q)}{2} \right) \delta_{\mu i}\Big]_{\alpha\beta}$$

$$+ \delta_6 \sqrt{8\pi^2\sigma} \left[\Sigma^{jk}\right]_{\alpha\beta} \left(-iq_j\right) \delta_{\mu k},$$



$$= \delta_V = -8\pi^2 \alpha\, \delta_2 \sqrt{\frac{d}{2(d-1)}} \delta_{\mu i} \delta_{\nu j} \left[\not{\chi}^{ij}\right]_{\alpha\beta} \tag{50}$$

It is clear that the counter terms are determined from the self-energy correction to the inverse propagator alone. The full renormalized perturbation series can now be set up in terms of these diagrams. Following standard technique we sum up all one-particle irreducible (1PI) diagrams [13] and find,



In terms of the vertex functions and the propagators the diagrams are evaluated to

$$\Sigma^{1\mathrm{L}}_{\psi}(k) = \int_q C^{\mu}(k+q,-q,-k) G(k+q) C^{\nu}(k,q,-k-q) D^{\mathrm{gf}}_{\mu\nu}(q)$$
$$+ \int_q V^{\mu\nu}(k,q,-q,-k) D^{\mathrm{gf}}_{\mu\nu}(q) + \delta_{\Sigma},$$

$$\Sigma_A^{\mu\nu,1\mathrm{L}}(k) = -\int_q \mathrm{tr}\left[C^\mu(q+k,-k,q)G(q+k)C^\nu(q,k,-q-k)G(q)\right] - \int_q \mathrm{tr}\left[V^{\mu\nu}(q,k,-k,-q)G(q)\right] + \delta_\Pi, \tag{51}$$

where

$$\int_q = \int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d} \tag{52}$$

is the shorthand for the $d+1$ dimensional integral over internal frequency-momenta q. For one insertion of Eq. (39) we replace an internal photon propagator,

$$D^{\mathrm{gf}}_{\mu\nu}(q) \mapsto D^{\mathrm{gf}}_{\mu\rho}(q)O_c^{\rho\sigma}(q)D^{\mathrm{gf}}_{\sigma\nu}(q), \tag{53}$$

and find in the fermion self-energy the additional term

$$\Sigma^{1\mathrm{L}}_{\psi,c}(k) = \int_q C^\mu(k+q,-q,-k)G(k+q)C^\nu(k,q,-k-q)D^{\mathrm{gf}}_{\mu\rho}(q)O_c^{\rho\sigma}(q)D^{\mathrm{gf}}_{\sigma\nu}(q) + \int_q V^{\mu\nu}(k,q,-q,-k)D^{\mathrm{gf}}_{\mu\rho}(q)O_c^{\rho\sigma}(q)D^{\mathrm{gf}}_{\sigma\nu}(q). \tag{54}$$

In the photon self-energy we find no internal photon line and hence no insertion at one loop; from its $F_{ij}F_{ij}$ part at $c=0$ we fix δ_4. To determine the renormalization parameters we shall demand the finiteness of the tree level action with respect to an artificial renormalization scale μ, which implies that the divergent pieces in the self energy expressions should identically cancel. This criterion shall determine the divergent counter terms of the action.

It can be shown that the only external momentum dependence of self-energies is through the set of functions $(k_0, g_a(k), g^i_a(k))$. Therefore, we may perform a Taylor expansion to obtain the leading order external momentum dependence of the self-energies in terms of these functions.

C. Useful formulae for the one-loop integrals

We shall compute the frequency part of the loop integrals exactly as they involve no divergences. For the remaining d dimensional momentum integral we shall first resolve the complicated denominators of the frequency-integrated loop expressions using Feynman's parametric integral formulae [15]:

$$\frac{1}{A^aB^b} = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\int_0^1 dx \frac{x^{a-1}(1-x)^{b-1}}{[Ax+B(1-x)]^{a+b}},$$
$$\frac{1}{A_1^{a_1}\dots A_n^{a_n}} = \frac{\Gamma(a_1+\dots+a_n)}{\Gamma(a_1)\dots\Gamma(a_n)}\int_0^1 dx_1\cdots\int_0^1 dx_n \frac{\delta(1-(x_1+\dots+x_n))x_1^{a_1-1}\dots x_n^{a_n-1}}{(x_1A_1+\dots+x_nA_n)^{a_1+\dots+a_n}}. \tag{55}$$

Following the Feynman parameterization, a momentum shift puts the generic loop integral in the form, $\int_q f_1(q^2)f_2(q_i,k_\mu)$, which can be further reduced by considering the integral identity,

$$\int \frac{d^dq}{(2\pi)^d} f(q^2)q_iq_j = \frac{\delta_{ij}}{d}\int \frac{d^dq}{(2\pi)^d} f(q^2)q^2. \tag{56}$$

In fact we shall need similar identities involving multiple products of internal momentum with open indices. In such integrals the only non-zero contribution comes from cases where all the internal momentum indices are pairwise contracted. It is straightforward to show this with the help of the Laplace transform of the (well-behaved) function $f(q^2)$. With the transform the integrals of the type,

$$K_{ijkl} = \int \frac{d^dq}{(2\pi)^d} f(q^2) q_i q_j q_k q_l, \tag{57}$$

$$= \int_0^\infty dt \int \frac{d^dq}{(2\pi)^d} \tilde{f}(t) e^{-q^2 t} q_i q_j q_k q_l, \tag{58}$$

can be recast in terms of a generating variable Y_i such that,

$$\begin{aligned} K_{ijkl} &= \frac{\partial^4}{\partial Y_i \partial Y_j \partial Y_k \partial Y_l} \left[\int_0^\infty dt \int \frac{d^dq}{(2\pi)^d} \tilde{f}(t) e^{-q^2 t + Y_i q_i} \right] \Big|_{Y_i \to 0}, \\ &= \frac{\partial^4}{\partial Y_i \partial Y_j \partial Y_k \partial Y_l} \left[\int_0^\infty dt \tilde{f}(t) \frac{e^{Y_i^2/(4t)}}{(2\pi)^{d/2} (2t)^{d/2}} \right] \Big|_{Y_i \to 0}, \\ &= \int_0^\infty dt \frac{\tilde{f}(t)}{(2\pi)^{d/2} (2t)^{d/2}} \left(\frac{\delta_{ij}}{2t} \frac{\delta_{kl}}{2t} + \frac{\delta_{ik}}{2t} \frac{\delta_{jl}}{2t} + \frac{\delta_{il}}{2t} \frac{\delta_{jk}}{2t} \right). \end{aligned} \tag{59}$$

On the other hand we find that,

$$\begin{aligned} \int \frac{d^dq}{(2\pi)^d} f(q^2) q^4 &= \int_0^\infty dt \int \frac{d^dq}{(2\pi)^d} \tilde{f}(t) e^{-q^2 t} q^4, \\ &= \int_0^\infty dt \tilde{f}(t) \frac{\partial^2}{\partial t^2} \int \frac{d^dq}{(2\pi)^d} e^{-q^2 t}, \\ &= \int_0^\infty dt \tilde{f}(t) \frac{\partial^2}{\partial t^2} \left(\frac{1}{(2\pi)^{d/2} (2t)^{d/2}} \right), \\ &= \int_0^\infty dt \frac{\tilde{f}(t)}{(2\pi)^{d/2} (2t)^{d/2}} \left(\frac{d(d+2)}{4t^2} \right). \end{aligned}$$

Combining these two results we find that,

$$K_{ijkl} = \int \frac{d^dq}{(2\pi)^d} f(q^2) q_i q_j q_k q_l = \frac{1}{d(d+2)} \int \frac{d^dq}{(2\pi)^d} f(q^2) q^4 \left(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} \right), \tag{60}$$

and similarly,

$$K_{ijklmn} = \int \frac{d^dq}{(2\pi)^d} f(q^2) q_i q_j q_k q_l q_m q_n = \frac{1}{d(d+2)(d+4)} \int \frac{d^dq}{(2\pi)^d} f(q^2) q^4 \left(\delta_{ij}\delta_{kl}\delta_{mn} + \dots \right). \tag{61}$$

[ToDo: Hunch: q^6 in place of q^4 under the integral on the right; cf. q^4 for four momenta in Eq. (60).]

To proceed with the computation in the presence of anisotropies we shall need a few integral identities that we borrow from Ref. [4],

$$(a) \int \frac{d^dq}{(2\pi)^d} f(q^2) q_i q_j = \frac{\delta_{ij}}{d} \int \frac{d^dq}{(2\pi)^d} f(q^2) q^2,$$

$$(b)\int\frac{d^dq}{(2\pi)^d}f(q^2)g_a(q)g_b(q)=\frac{2\delta_{ab}}{(d-1)(d+2)}\int\frac{d^dq}{(2\pi)^d}f(q^2)q^4,$$

$$(c)\int\frac{d^dq}{(2\pi)^d}f(q^2)g_a(q)g_b(q)g_c(q)g_d(q)=\left(\frac{d}{2(d-1)}\right)^2\frac{16}{d(d+2)(d+4)(d+6)}\int\frac{d^dq}{(2\pi)^d}f(q^2)q^8$$
$$\times\left(\delta_{ab}\delta_{cd}+\delta_{ac}\delta_{bd}+\delta_{ad}\delta_{bc}+\mathrm{tr}\left[\lambda_a\lambda_b\lambda_c\lambda_d\right]+\mathrm{tr}\left[\lambda_b\lambda_a\lambda_c\lambda_d\right]+\mathrm{tr}\left[\lambda_b\lambda_c\lambda_a\lambda_d\right]\right). \tag{62}$$

These identities can be derived from the symmetric properties of the Gell-Mann matrices. For example, we can show that the integral

$$\int\frac{d^dq}{(2\pi)^d}f(q^2)g_a(q)g_b(q)=\frac{d}{2(d-1)}\int\frac{d^dq}{(2\pi)^d}f(q^2)\lambda_a^{ij}\lambda_b^{kl}q_iq_jq_kq_l, \tag{63}$$

vanishes identically unless the indices of the product $q_iq_jq_kq_l$ are pairwise contracted, by the Laplace transform argument given above for Eq. (60). Using these identities we can further derive the contractions within the integrals that arise in our computations.

$$(a)\int\frac{d^dq}{(2\pi)^d}f(q^2)g_a^i(q)g_b^i(q)=\frac{d}{2(d-1)}\int\frac{d^dq}{(2\pi)^d}f(q^2)\lambda_a^{ij}\lambda_b^{ik}q_jq_k=\frac{\delta_{ab}}{(d-1)}\int\frac{d^dq}{(2\pi)^d}f(q^2)q^2,$$

$$(b)\int\frac{d^dq}{(2\pi)^d}f(q^2)g_a(q)g_b^i(q)g_c(q)g_d^i(q)$$
$$=\left(\frac{d}{2(d-1)}\right)^2\int\frac{d^dq}{(2\pi)^d}f(q^2)\lambda_a^{jk}\lambda_b^{il}\lambda_c^{mn}\lambda_d^{ip}(q_jq_k)q_l(q_mq_n)q_p$$
$$=\frac{d}{4(d-1)^2(d+2)(d+4)}\int\frac{d^dq}{(2\pi)^d}f(q^2)q^6\lambda_a^{jk}\lambda_b^{il}\lambda_c^{mn}\lambda_d^{ip}$$
$$\times\Bigg(\delta_{lp}\left(\delta_{jm}\delta_{kn}+\delta_{jn}\delta_{km}\right)+\delta_{lj}\left(\delta_{km}\delta_{pn}+\delta_{kn}\delta_{pm}\right)+\delta_{lk}\left(\delta_{jm}\delta_{pn}+\delta_{jn}\delta_{pm}\right)$$
$$+\delta_{lm}\left(\delta_{jn}\delta_{kp}+\delta_{jp}\delta_{kn}\right)+\delta_{ln}\left(\delta_{jm}\delta_{kp}+\delta_{jp}\delta_{km}\right)\Bigg)$$
$$=\frac{d}{2(d-1)^2(d+2)(d+4)}\int\frac{d^dq}{(2\pi)^d}f(q^2)q^6\Bigg(\mathrm{tr}\left[\lambda_a\lambda_c\right]\mathrm{tr}\left[\lambda_b\lambda_d\right]+2\,\mathrm{tr}\left[\lambda_b\lambda_a\lambda_c\lambda_d\right]$$
$$+2\,\mathrm{tr}\left[\lambda_b\lambda_c\lambda_a\lambda_d\right]\Bigg)$$

We shall also need a formula for the contraction $s_a\lambda_a^{ij}\lambda_a^{kl}$. We have,

$$s_a=\begin{cases}1\ \forall\ a\in\text{off-diag}\\ -1\ \forall\ a\in\text{diag}\end{cases}, \tag{64}$$

where off-diag and diag correspond to the indices of the off-diagonal and the diagonal λ matrices respectively. We already know,

$$\lambda_a^{ij}\lambda_a^{kl}=\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk}-\frac{2}{d}\delta_{ij}\delta_{kl}. \tag{65}$$

Furthermore, the d dimensional off-diagonal λ matrices are parametrized by the formula [3]

$$\lambda^{jk}_{(rs)} = \delta_{rj}\delta_{sk} + \delta_{rk}\delta_{sj} \;\forall\; (rs) \mid 1 \leq s < r \leq d. \tag{66}$$

We can easily show that for the off-diagonal lambda matrices,

$$\sum_{a\in\text{off-diag}} \lambda^{ij}_a \lambda^{kl}_a = \sum_{r>s}^{d}\sum_{s=1}^{d} \lambda^{ij}_{(rs)}\lambda^{kl}_{(rs)} \tag{67}$$

$$= \sum_{r>s}^{d}\sum_{s=1}^{d} \left(\delta_{ri}\delta_{sj} + \delta_{rj}\delta_{si}\right)\left(\delta_{rk}\delta_{sl} + \delta_{rl}\delta_{sk}\right) \tag{68}$$

$$= \sum_{r>s}^{d}\sum_{s=1}^{d} \left[\delta_{ri}\delta_{sj}\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right) + \delta_{rj}\delta_{si}\left(\delta_{jk}\delta_{il} + \delta_{jl}\delta_{ik}\right)\right] \tag{69}$$

$$= \sum_{r>s}^{d}\sum_{s=1}^{d} \delta_{ri}\delta_{sj}\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right) + \sum_{s>r}^{d}\sum_{r=1}^{d} \delta_{sj}\delta_{ri}\left(\delta_{jk}\delta_{il} + \delta_{jl}\delta_{ik}\right) \tag{70}$$

$$= \sum_{r=1}^{d}\sum_{s\neq r}^{d} \delta_{ri}\delta_{sj}\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right) \tag{71}$$

$$= \left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right) - 2\delta_{ij}\delta_{kl}\delta_{ik}. \tag{72}$$

We are now able to derive the desired formula,

$$\sum_a s_a \lambda^{ij}_a \lambda^{kl}_a = -\sum_{a\in\text{diag}} \lambda^{ij}_a \lambda^{kl}_a + \sum_{a\in\text{off-diag}} \lambda^{ij}_a \lambda^{kl}_a \tag{73}$$

$$= -\sum_a \lambda^{ij}_a \lambda^{kl}_a + 2\sum_{a\in\text{off-diag}} \lambda^{ij}_a \lambda^{kl}_a \tag{74}$$

$$= -\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \frac{2}{d}\delta_{ij}\delta_{kl}\right) + 2\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - 2\delta_{ij}\delta_{kl}\delta_{ik}\right) \tag{75}$$

$$= \left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right) + \frac{2}{d}\delta_{ij}\delta_{kl} - 4\delta_{ij}\delta_{kl}\delta_{ik}. \tag{76}$$

In our case, quite generally, various loop integrals depend on the external momentum k through the functions $g_a(k+q')$ and $g^i_a(k+q'')$ where q' and q'' are internal loop momenta. We can combine them together and say that a generic loop integral is a function $f(X_\alpha)$ with $X_\alpha(k) = \left\{g_a(k), g^i_a(k)\right\}$. We can further define a parametric linear differential operator,

$$\Delta_{ij}|_{X(k)} = \left(\frac{\partial^2}{\partial k_i \partial k_j} - \sum_{\alpha,\beta} \frac{\partial X_\alpha}{\partial k_i}\frac{\partial X_\beta}{\partial k_j}\frac{\partial^2}{\partial X_\alpha \partial X_\beta}\right), \tag{77}$$

which obeys the Leibniz rule,

$$\Delta_{ij}\left(u(X)v(X)\right) = \left(\frac{\partial^2 (uv)}{\partial k_i \partial k_j} - \sum_{\alpha,\beta} \frac{\partial X_\alpha}{\partial k_i}\frac{\partial X_\beta}{\partial k_j}\frac{\partial^2 (uv)}{\partial X_\alpha \partial X_\beta}\right),$$

$$
\begin{aligned}
=&\Bigg(u\frac{\partial^2 v}{\partial k_i \partial k_j} + v\frac{\partial^2 u}{\partial k_i \partial k_j} + \cancel{\frac{\partial u}{\partial k_i}\frac{\partial v}{\partial k_j}} + \cancel{\frac{\partial v}{\partial k_i}\frac{\partial u}{\partial k_j}} \\
&- \sum_{\alpha,\beta} \frac{\partial X_\alpha}{\partial k_i}\frac{\partial X_\beta}{\partial k_j}\left(u\frac{\partial^2 v}{\partial X_\alpha \partial X_\beta} + v\frac{\partial^2 u}{\partial X_\alpha \partial X_\beta} + \cancel{\frac{\partial u}{\partial X_\alpha}\frac{\partial v}{\partial X_\beta}} + \cancel{\frac{\partial v}{\partial X_\alpha}\frac{\partial u}{\partial X_\beta}}\right)\Bigg), \\
=&u(X)\Delta_{ij}(v(X)) + v(X)\Delta_{ij}(u(X)). \qquad (78)
\end{aligned}
$$

We strike the terms that cancel each other, since $\sum_\alpha \frac{\partial X_\alpha}{\partial k_i}\frac{\partial u}{\partial X_\alpha} = \frac{\partial u}{\partial k_i}$. For functions of variables $X_\alpha(k)$, the differential operator can alternatively be expressed as,

$$
\Delta_{ij}(f(X_\alpha)) = \sum_\alpha \frac{\partial^2 X_\alpha}{\partial k_i \partial k_j}\frac{\partial f}{\partial X_\alpha}, \qquad (79)
$$

which follows from applying the chain rule of derivatives on its definition. On functions of X, Δ_{ij} is thus the first-order operator $\sum_\alpha \frac{\partial^2 X_\alpha}{\partial k_i \partial k_j}\frac{\partial}{\partial X_\alpha}$, the Lie derivative along this vector field, and the Leibniz rule above is its derivation property. We shall define another differential operator,

$$
\Delta_a = \frac{1}{4}\sqrt{\frac{2(d-1)}{d}}\sum_{i,j}\lambda_a^{ij}\Delta_{ij}, \qquad (80)
$$

which acts as a partial derivative of the band momentum $g_a(k)$. Some simple properties can be immediately verified:

$$
\begin{aligned}
\Delta_a(g_b(k+q')) =&\frac{1}{4}\left(\lambda_a^{ij}\frac{\partial^2\left(\lambda_c^{mn}k_m k_n\right)}{\partial k_i \partial k_j}\frac{\partial g_b(k+q')}{\partial g_c(k)} + \lambda_a^{ij}\frac{\partial^2\left(\lambda_c^{mn}k_m\right)}{\partial k_i \partial k_j}\frac{\partial g_b(k+q')}{\partial g_c^n(k)}\right), \\
=&\frac{1}{2}\operatorname{tr}\left[\lambda_a\lambda_b\right] = \delta_{ab}, \\
\Delta_a(g_b^i(k+q')) =&\frac{1}{4}\lambda_a^{ij}\frac{\partial^2\left(\lambda_c^{mn}k_m\right)}{\partial k_i \partial k_j}\frac{\partial g_b^i(k+q')}{\partial g_c^n(k)} = 0, \\
\Delta_a\left(f(g_a(k+q'), g_a^i(k+q''))\right) =&\frac{1}{4}\lambda_a^{ij}\frac{\partial^2\left(\lambda_b^{mn}k_m k_n\right)}{\partial k_i \partial k_j}\frac{\partial f}{\partial g_b(k)} + \frac{1}{4}\lambda_a^{ij}\frac{\partial^2\left(\lambda_b^{mn}k_m\right)}{\partial k_i \partial k_j}\frac{\partial f}{\partial g_b^n(k)}, \\
=&\frac{\partial f}{\partial g_a(k)}, \qquad (81) \\
\Delta_a^2\left(f(g_a(k+q'), g_a^i(k+q''))\right) =&\frac{\partial^2 f}{\partial g_a(k)^2}, \qquad (82)
\end{aligned}
$$

where we have made use of the formulae,

$$
\begin{aligned}
g_a(k+q) &= \sqrt{\frac{d}{2(d-1)}}\lambda_a^{ij}(k_i+q_i)(k_j+q_j) = g_a(k) + 2g_a^i(k)q_i + g_a(q), \\
g_a^i(k+q) &= g_a^i(k) + g_a^i(q). \qquad (83)
\end{aligned}
$$

We can abbreviate, $\Delta^2 = \sum_a \Delta_a^2$. We also find that,

$$
\Delta^2 f(g_a(k+q'), g_a^i(k+q'')) = \sum_a \frac{\partial^2 f(g_a(k+q'), g_a^i(k+q''))}{\partial g_a(k)^2}. \qquad (84)
$$

It also follows that,

$$\Delta^2 k^4 = \Delta^2 \sum_a g_a(k)^2 = (d-1)(d+2). \tag{85}$$

D. The fermion self-energy

We start with the one-loop renormalized fermion self-energy $\Sigma_\psi^{1\mathrm{L}}(k)$. The full expression is,

$$\begin{aligned}\Sigma_\psi^{1\mathrm{L}}(k) =& 8\pi^2\alpha \int_q \Bigg[\left[i\delta_{\mu 0} + \frac{\not{q}^i(q+2k)}{2}\delta_{\mu i} \right] \\ &\times \left[\frac{-i(k_0+q_0) + \not{q}(k+q)/2}{(k_0+q_0)^2 + (k+q)^4/4} \right] \\ &\times \left[i\delta_{\nu 0} + \frac{\not{q}^i(q+2k)}{2}\delta_{\nu i} \right] \\ &\times \left(\delta_{\mu 0}\delta_{\nu 0}\frac{\rho q^2}{q_0^2+\rho q^4} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+\rho q^4} \right) \Bigg] - \left[\delta_1 \left(ik_0\right) + \delta_2 \left(\frac{\not{q}(k)}{2} \right) \right]. \end{aligned} \tag{86}$$

The contribution from the quartic vertex identically vanishes in this case. We shall consider the case for non-zero external frequency and momenta separately. For $k = (k_0, 0)$ we obtain,

$$\begin{aligned}\mathrm{tr}\left[\Sigma_\psi^{1\mathrm{L}}(k_0,0)\right] =& 8\pi^2\alpha \int_q \mathrm{tr} \left[\left[i\delta_{\mu 0} + \frac{\not{q}^i(q)}{2}\delta_{\mu i} \right] \left[\frac{-i(q_0+k_0) + \not{q}(q)/2}{(q_0+k_0)^2 + q^4/4} \right] \right. \\ &\left. \times \left[i\delta_{\nu 0} + \frac{\not{q}^i(q)}{2}\delta_{\nu i} \right] \left(\delta_{\mu 0}\delta_{\nu 0}\frac{\rho q^2}{q_0^2+\rho q^4} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+\rho q^4} \right) \right] - 4\delta_1\left(ik_0\right), \\ =& -4i\pi^2\alpha\left(d - 8\rho + 2\right) \int_q \frac{(q_0+k_0)q^2}{(q_0^2+\rho q^4)((q_0+k_0)^2 + \frac{q^4}{4})} - 4\delta_1(ik_0). \end{aligned} \tag{87}$$

[ToDo: Hunch: $4 = \mathrm{tr}_1$ **at** $d = 3$ **in** $-4\delta_1(ik_0)$**; for general** d**,** $\mathrm{tr}_1 = 2^{\lfloor d_\gamma/2 \rfloor}$**.]**

E. The photon self-energy

Now we turn to the counter terms in the photon sector. It shall be convenient to introduce the transverse and longitudinal projectors associated with the photon propagator, defined respectively as,

$$P^T_{\mu\nu}(k) = \delta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2},\ P^L_{\mu\nu}(k) = \frac{k_\mu k_\nu}{k^2}. \tag{88}$$

We shall also define similar but separate projectors that act on the spatial components alone,

$$p^T_{ij}(k) = \delta_{ij} - \frac{k_i k_j}{k^2},\ p^L_{ij}(k) = \frac{k_i k_j}{k^2}. \tag{89}$$

With the aid of these projectors the one-loop renormalized photon self-energy can be written down as,

$$\begin{aligned}\Sigma_A^{\mu\nu,1\mathrm{L}}(k_0,\vec{k}) =& -8\pi^2\alpha\int_q \mathrm{tr}\left[\left[i\delta_{\mu 0}+\frac{\delta_{\mu i}\not{g}^i(k+2q)}{2}\right]\right.\\ &\times\left[\frac{-i(q_0+k_0)+\frac{\not{g}(q+k)}{2}}{(q_0+k_0)^2+\frac{|\vec{q}+\vec{k}|^4}{4}}\right]\\ &\left.\times\left[i\delta_{\nu 0}+\frac{\delta_{\nu i}\not{g}^i(k+2q)}{2}\right]\left[\frac{-iq_0+\frac{\not{g}(q)}{2}}{q_0^2+\frac{q^4}{4}}\right]\right]\\ &-\delta_3\left(k^2P^T_{\mu\nu}(k)-k^2\delta_{\mu i}\delta_{\nu j}p^T_{ij}(k)\right)\\ &-\delta_5\delta_{\mu i}\delta_{\nu j}\rho k^4 p^T_{ij}(k).\end{aligned} \tag{90}$$

The contribution from the quartic vertex once again drops out identically due to the properties of the λ_a matrices. The computation also greatly simplifies if we evaluate the self-energy at zero external frequency, $k_0 = 0$.

To extract the counter term δ_5 we separate the spatial components of the photon self-energy and evaluate it at the fixed external frequency $k_0 = 0$. We obtain,

$$\begin{aligned}&\Sigma_A^{ij,1\mathrm{L}}(0,\vec{k})\\ =& -8\pi^2\alpha\int_q \mathrm{tr}\left[\left[\frac{\not{g}^i(k+2q)}{2}\right]\left[\frac{-iq_0+\frac{\not{g}(q+k)}{2}}{q_0^2+\frac{|\vec{q}+\vec{k}|^4}{4}}\right]\right.\\ &\left.\times\left[\frac{\not{g}^j(k+2q)}{2}\right]\left[\frac{-iq_0+\frac{\not{g}(q)}{2}}{q_0^2+\frac{q^4}{4}}\right]\right]-\delta_5\rho k^4 p^T_{ij}(k),\end{aligned} \tag{91}$$

$$= -2\pi^2\alpha\int_q \frac{M_1^{ij}(\vec{k},q_0,\vec{q})}{\left(q_0^2+\frac{|\vec{q}+\vec{k}|^4}{4}\right)\left(q_0^2+\frac{q^4}{4}\right)}-\delta_5\rho k^4 p^T_{ij}(k), \tag{92}$$

where,

$$M_1^{ij}(\vec{k},q_0,\vec{q}) = \mathrm{tr}\left[\not{g}^i(k+2q)\left(-iq_0+\frac{\not{g}(q+k)}{2}\right)\not{g}^j(k+2q)\left(-iq_0+\frac{\not{g}(q)}{2}\right)\right], \tag{93}$$

is the trace product in the numerator. We compute the frequency integral exactly using the residue theorem and obtain,

$$\begin{aligned}&\Sigma_A^{ij,1\mathrm{L}}(0,\vec{k})\\ =& -2\pi^2\alpha\int_{\vec{q}}\left(-\frac{1}{2}\frac{M_1^{ij}(\vec{k},i|\vec{k}+\vec{q}|^2/2,\vec{q})}{\frac{|\vec{k}+\vec{q}|^2}{2}\left(\frac{|\vec{k}+\vec{q}|^2}{2}-\frac{q^2}{2}\right)\left(\frac{|\vec{k}+\vec{q}|^2}{2}+\frac{q^2}{2}\right)}+\frac{1}{2}\frac{M_1^{ij}(\vec{k},iq^2/2,\vec{q})}{\frac{q^2}{2}\left(\frac{|\vec{k}+\vec{q}|^2}{2}-\frac{q^2}{2}\right)\left(\frac{|\vec{k}+\vec{q}|^2}{2}+\frac{q^2}{2}\right)}\right)\\ &-\delta_5\rho k^4 p^T_{ij}(k),\\ =&\frac{8\pi^2\alpha\, d}{4(d-1)^2}\int_{\vec{q}}\frac{M_2^{ij}(\vec{k},\vec{q})}{q^2\left(q^2+2\vec{q}\cdot\vec{k}+k^2\right)\left(q^2+\frac{2\vec{q}\cdot\vec{k}+k^2}{2}\right)}\end{aligned}$$

$$
\begin{aligned}
=&\frac{16\pi^2\alpha\, d}{4(d-1)^2}\int_{\vec{q}}\int_0^1 dx_1\int_0^1 dx_2\int_0^1 dx_3\frac{\delta\left(1-(x_1+x_2+x_3)\right)M_2^{ij}(\vec{k},\vec{q})}{\left(x_1q^2+x_2\left(q^2+2\vec{q}\cdot\vec{k}+k^2\right)+x_3\left(q^2+\frac{2\vec{q}\cdot\vec{k}+k^2}{2}\right)\right)^3},\\
=&\frac{16\pi^2\alpha\, d}{4(d-1)^2}\int_{\vec{q}}\int_0^1 dx_1\int_0^1 dx_2\int_0^1 dx_3\\
&\times\frac{\delta\left(1-(x_1+x_2+x_3)\right)M_2^{ij}(\vec{k},\vec{q})}{\left(q^2+2\vec{q}\cdot\left(x_2\vec{k}+(x_3/2)\vec{k}\right)+\left|x_2\vec{k}+(x_3/2)\vec{k}\right|^2+(x_2+x_3/2)\left(1-(x_2+x_3/2)\right)k^2\right)^3},\\
=&\frac{16\pi^2\alpha\, d}{4(d-1)^2}\int_{\vec{q}}\int_0^1 dx_1\int_0^1 dx_2\int_0^1 dx_3\frac{\delta\left(1-(x_1+x_2+x_3)\right)M_2^{ij}(\vec{k},\vec{q}-(x_2+x_3/2)\vec{k})}{\left(q^2+(x_2+x_3/2)\left(1-(x_2+x_3/2)\right)k^2\right)^3},
\end{aligned}
$$

where in the last two lines we have used the delta function condition $x_1+x_2+x_3=1$ and shifted the internal momenta to simplify the integral.

III. DIMENSIONAL REGULARIZATION

Here we continue the loop momenta to $d=4-\epsilon$ and find the divergences as poles in ϵ in place of the shell.

To determine the renormalization parameters we need to define the renormalization conditions that leave the tree level action unaffected,

$$
\begin{aligned}
\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\left[\partial_{k_0}\Sigma(k_0,0)\right]|_{k_0\to 0}&=0,\\
\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\left[\not{\Delta}\Sigma(0,\vec{k})\right]\Big|_{\vec{k}\to 0}&=0,\\
\partial_{k^2}\Pi^{00}(0,\vec{k})\Big|_{\vec{k}\to 0}&=0,\\
\Delta^2\Pi^{ii}(0,\vec{k})\Big|_{\vec{k}\to 0}&=0,
\end{aligned}
\tag{94}
$$

where $\not{\Delta}=\sum_a\gamma^a\Delta_a|_{\{g_a(k),g_a^i(k)\}}$ is the differential operator defined above. These renormalization conditions alone determine the renormalization parameters of the problem.

A. The fermion self-energy

We start with the one-loop renormalized fermion self-energy $\Sigma^{1\mathrm{L}}(k)$. The full expression is,

$$
\begin{aligned}
\Sigma^{1\mathrm{L}}(k)=\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\Bigg[&e\big[i\delta_{\mu 0}+\frac{-\not{g}^i(q)+2\not{g}^i(k)}{2m}\delta_{\mu i}\big]\\
&\times\left[\frac{-i(k_0-q_0)+\not{g}(k-q)/(2m)}{(k_0-q_0)^2+(k-q)^4/(4m^2)}\right]\\
&\times e\big[i\delta_{\nu 0}+\frac{-\not{g}^i(q)+2\not{g}^i(k)}{2m}\delta_{\nu i}\big]
\end{aligned}
$$

$$\times \left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)$$
$$+\left[\frac{\sqrt{\frac{d}{2(d-1)}}\lambda_a^{ij}\delta_{\mu i}\delta_{\nu j}\gamma^a}{2m}\right]\left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)\Bigg]$$
$$-\left[\delta_1\left(ik_0\right)+\delta_2\left(\frac{\not{g}(k)}{2m}\right)\right]. \tag{95}$$

The expression simplifies because the contribution from the quartic vertex vanishes identically by the property $\lambda_a^{ii}=0$,

$$\left[\frac{\sqrt{\frac{d}{2(d-1)}}\lambda_a^{ij}\delta_{\mu i}\delta_{\nu j}\gamma^a}{2m}\right]\left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)=0. \tag{96}$$

The remainder of the expression becomes,

$$\Sigma^{1\mathrm{L}}(k)=e^2\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\Bigg[-\left[\frac{-i(k_0-q_0)+\not{g}(k-q)/(2m)}{(k_0-q_0)^2+(k-q)^4/(4m^2)}\right]\frac{v^2q^2}{q_0^2+v^2q^4}$$
$$+\left[\frac{\not{g}^i(-q+2k)\left(-i(k_0-q_0)+\not{g}(k-q)/(2m)\right)\not{g}^i(-q+2k)}{(k_0-q_0)^2+(k-q)^4/(4m^2)}\right]\frac{1}{q_0^2+v^2q^4}\Bigg]$$
$$-\left[\delta_1\left(ik_0\right)+\delta_2\left(\frac{\not{g}(k)}{2m}\right)\right]. \tag{97}$$

Further simplification is afforded when we consider fixed values of external frequency and momentum as required by the renormalization conditions. We choose the fixed external momenta suitable for the first renormalization condition and find,

$$\Sigma^{1\mathrm{L}}(k_0,0)=\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\Bigg[e\Big[i\delta_{\mu 0}-\frac{\not{g}^i(q)}{2m}\delta_{\mu i}\Big]$$
$$\times\left[\frac{-i(k_0-q_0)+\not{g}(q)}{(k_0-q_0)^2+q^4/(4m^2)}\right]$$
$$\times e\Big[i\delta_{\nu 0}-\frac{\not{g}^i(q)}{2m}\delta_{\nu i}\Big]$$
$$\times\left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)\Bigg]-\delta_1\left(ik_0\right). \tag{98}$$

[ToDo: Hunch: $-i(k_0-q_0)+\not{g}(q)/(2m)$ in the numerator above, as in the next equation.] We find that the k_0 derivative of the fermion propagator gives,

$$\lim_{k_0\to 0}\partial_{k_0}\left[\frac{-i(k_0-q_0)+\not{g}(q)/(2m)}{(k_0-q_0)^2+q^4/(4m^2)}\right]\Bigg|_{k_0\to 0}=\frac{i\left(q_0^2-\frac{q^4}{4m^2}\right)+2q_0\not{g}(q)/(2m)}{\left(q_0^2+\frac{q^4}{4m^2}\right)^2}. \tag{99}$$

We resolve the matrix numerator of the self-energy expression,

$$\begin{aligned}
&\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\left[\left[i\delta_{\mu 0}-\frac{\not{g}^i(q)}{2m}\delta_{\mu i}\right]\left[i\left(q_0^2-\frac{q^4}{4m^2}\right)+\frac{2q_0\not{g}(q)}{2m}\right]\left[i\delta_{\nu 0}-\frac{\not{g}^i(q)}{2m}\delta_{\nu i}\right]\right]\\
=&i\left(q_0^2-\frac{q^4}{4m^2}\right)\left(i\delta_{\mu 0}i\delta_{\nu 0}+\frac{g_a^i(q)g_a^j(q)}{4m^2}\delta_{\mu i}\delta_{\nu j}\right)-\frac{2q_0 g_a(q)}{2m}\left(\frac{g_a^i(q)}{2m}i\delta_{\mu i}\delta_{\nu 0}+\frac{g_a^i(q)}{2m}i\delta_{\nu i}\delta_{\mu 0}\right). \quad (100)
\end{aligned}$$

Leaving out the odd portion of the integral which vanishes identically we have for the first of the renormalization conditions (94),

$$\begin{aligned}
&\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\partial_{k_0}\,\mathrm{tr}\left[\Sigma^{1\mathrm{L}}(k_0,0)\right]_{k_0\to 0}\\
=&e^2\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\left[\frac{i\left(q_0^2-\frac{q^4}{4m^2}\right)}{\left(q_0^2+\frac{q^4}{4m^2}\right)^2}\left(i\delta_{\mu 0}i\delta_{\nu 0}+\frac{g_a^k(q)g_a^l(q)}{4m^2}\delta_{\mu k}\delta_{\nu l}\right)\right.\\
&\left.\times\left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)\right]-i\delta_\psi,\\
=&ie^2\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\frac{\left(q_0^2-\frac{q^4}{4m^2}\right)}{\left(q_0^2+\frac{q^4}{4m^2}\right)^2}\left[-\frac{v^2q^2}{q_0^2+v^2q^4}+\frac{d+2}{8m^2}\frac{q^2}{q_0^2+v^2q^4}\right]-i\delta_1,\\
=&\frac{ie^2}{v}\left(\frac{8-\frac{d+2}{m^2v^2}}{4(2+\frac{1}{mv})^2}\right)\int\frac{d^dq}{(2\pi)^d}\frac{1}{q^4}-i\delta_1. \quad (101)
\end{aligned}$$

[ToDo: Hunch: $\delta_\psi=\delta_1$ in the first line.] We shall compute the remaining momentum loop integral with dimensional regularization, i.e. we shall integrate over a fractional dimension $d=4-\epsilon$. To this end we shall include an artificial infrared regulator r^2 at the denominator of the integral and introduce a momentum scale μ because certain couplings such as $e^2=\tilde{e}^2\mu^{2+z_0-d}=\tilde{e}^2\mu^\epsilon$, are no longer dimensionless in the fractional dimension. In terms of the dimensionless parameters the counterterm correction becomes,

$$\begin{aligned}
\delta_1&=\frac{\tilde{e}^2}{\tilde{v}}\left(\frac{8-\frac{d+2}{\tilde{m}^2\tilde{v}^2}}{4(2+\frac{1}{\tilde{m}\tilde{v}})^2}\right)\mu^{2-z_0}\mu^{2+z_0-d}\int\frac{d^dq}{(2\pi)^d}\frac{1}{\left(q^2+r^2\right)^2}\\
&=\frac{\tilde{e}^2}{\tilde{v}}\left(\frac{8-\frac{d+2}{\tilde{m}^2\tilde{v}^2}}{4(2+\frac{1}{\tilde{m}\tilde{v}})^2}\right)\frac{\mu^{4-d}}{(4\pi)^{d/2}}\frac{\Gamma(2-d/2)}{\Gamma(2)}\left(r^2\right)^{d/2-2}, \quad (102)
\end{aligned}$$

where we have used the formula [15],

$$I(d,a,r;p)=\int\frac{d^dq}{(2\pi)^d}\frac{1}{(q^2+2\vec{q}\cdot\vec{p}+r^2)^a}=\frac{1}{(4\pi)^{d/2}}\frac{\Gamma(a-d/2)}{\Gamma(a)}\left(r^2-p^2\right)^{d/2-a}. \quad (103)$$

At this point we shall introduce two new dimensionless couplings,

$$\alpha=\frac{\tilde{e}^2\tilde{m}}{8\pi^2},$$

$$\rho = \tilde{m}^2\tilde{v}^2, \tag{104}$$

such that the prefactor of the integral becomes,

$$\frac{\tilde{e}^2}{\tilde{v}}\left(\frac{8-\frac{d+2}{\tilde{m}^2\tilde{v}^2}}{4(2+\frac{1}{\tilde{m}\tilde{v}})^2}\right) = 8\pi^2\alpha f_1(1/\sqrt{\rho}), \tag{105}$$

with,

$$f_1(x) = \frac{8x-(d+2)x^3}{4(2+x)^2}. \tag{106}$$

We can expand the expression for the counter term in small ϵ with $d = 4-\epsilon$, and find,

$$\begin{aligned}\delta_1 &= 8\pi^2\alpha f_1(1/\sqrt{\rho})\frac{\mu^{4-d}}{(4\pi)^{d/2}}\frac{\Gamma(2-d/2)}{\Gamma(2)}\left(r^2\right)^{d/2-2}\\ &= 8\pi^2\alpha f_1(1/\sqrt{\rho})\left(\frac{1}{8\pi^2\epsilon}+\frac{1}{16\pi^2}\left(\log\frac{4\pi\mu^2}{r^2}-\gamma\right)+O(\epsilon)\right),\end{aligned} \tag{107}$$

where γ is the Euler-Mascheroni constant. In what follows we shall adopt the minimal subtraction (MS) scheme to eliminate the $1/\epsilon$ pole singularity. This amounts to choosing the counter terms such that they only cancel the divergent part of the loop expansions [6, 15]. In this case

$$\delta_1 = 8\pi^2\alpha f_1(1/\sqrt{\rho})\left(\frac{1}{8\pi^2\epsilon}\right). \tag{108}$$

In the MS scheme the counter terms do not have an explicit dependence on the renormalization scale μ [15]. In our example this can be rationalized by absorbing the finite scale dependent part of the above expressions into the dimensionless couplings which are scale dependent by definition.

B. The band mass counter term

For the second renormalization condition it is preferable to consider the self energy correction with zero external frequency,

$$\begin{aligned}\Sigma^{1\mathrm{L}}(0,\vec{k}) = \int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\Bigg[&e\Big[i\delta_{\mu 0}+\frac{-\not{g}^i(q)+2\not{g}^i(k)}{2m}\delta_{\mu i}\Big]\\ &\times\left[\frac{iq_0+\not{g}(k-q)/(2m)}{q_0^2+(k-q)^4/(4m^2)}\right]\\ &\times e\Big[i\delta_{\nu 0}+\frac{-\not{g}^i(q)+2\not{g}^i(k)}{2m}\delta_{\nu i}\Big]\\ &\times\left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)\Bigg]-\delta_2\left[\frac{\not{g}(k)}{2m}\right],\\ =e^2\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}&\left[\left(\frac{1}{q_0^2+\frac{(k-q)^4}{4m^2}}\right)\right.\end{aligned}$$

$$\times \left[-\delta_{\mu 0}\delta_{\nu 0}\frac{\not{g}(k-q)}{2m} + \frac{\not{g}^k(-q+2k)\not{g}(k-q)\not{g}^l(-q+2k)}{8m^3}\delta_{\mu k}\delta_{\nu l} \right]$$
$$\times \left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4} \right) \Bigg] - \delta_2 \left[\frac{\not{g}(k)}{2m} \right]. \tag{109}$$

We can now take the trace-derivative as intended and find,

$$\begin{aligned}
&\frac{1}{2^{\lfloor d_\gamma/2 \rfloor}} \operatorname{tr}\left[\not{\Delta}\Sigma^{1\mathrm{L}}(0,\vec{k}) \right]\Big|_{\vec{k}\to 0} \\
&= \lim_{\vec{k}\to 0} e^2 \int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\frac{1}{2^{\lfloor d_\gamma/2 \rfloor}} \operatorname{tr}\Bigg[\left(\frac{1}{q_0^2 + \frac{|\vec{k}-\vec{q}|^4}{4m^2}} \right) \\
&\quad \times \left[-\delta_{\mu 0}\delta_{\nu 0}\frac{\gamma^a\gamma^a}{2m} + \frac{\left(-d_b^\mu(q)+2d_b^\mu(k)\right)\left(-d_c^\nu(q)+2d_c^\nu(k)\right)}{8m^3}\gamma^a\gamma^b\gamma^a\gamma^c \right] \\
&\quad \times \left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4} \right) \\
&\quad - \left(\frac{\frac{g_a(k-q)\gamma^a}{2m^2}}{\left(q_0^2 + \frac{|\vec{k}-\vec{q}|^4}{4m^2}\right)^2} \right) \\
&\quad \times \left[-\delta_{\mu 0}\delta_{\nu 0}\frac{g_a(k-q)\gamma^a}{2m} + \frac{\left(-d_a^\mu(q)+2d_a^\mu(k)\right)g_b(k-q)\left(-d_c^\nu(q)+2d_c^\nu(k)\right)}{8m^3}\gamma^a\gamma^b\gamma^c \right] \\
&\quad \times \left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4} \right) \Bigg] - \delta_2 \frac{1}{2^{\lfloor d_\gamma/2 \rfloor}} \operatorname{tr}\left[\frac{\gamma^a\gamma^a}{2m} \right], \\
&= e^2 \int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\frac{1}{2^{\lfloor d_\gamma/2 \rfloor}} \operatorname{tr}\Bigg[\left(\frac{1}{q_0^2 + \frac{\vec{q}^4}{4m^2}} \right) \\
&\quad \times \left[-\delta_{\mu 0}\delta_{\nu 0}\frac{\gamma^a\gamma^a}{2m} + \frac{d_b^\mu(q)d_c^\nu(q)}{8m^3}\gamma^a\gamma^b\gamma^a\gamma^c \right] \\
&\quad \times \left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4} \right) \\
&\quad - \left(\frac{\frac{g_a(q)\gamma^a}{2m^2}}{\left(q_0^2 + \frac{\vec{q}^4}{4m^2}\right)^2} \right) \\
&\quad \times \left[-\delta_{\mu 0}\delta_{\nu 0}\frac{g_a(q)\gamma^a}{2m} + \frac{d_a^\mu(q)g_b(q)d_c^\nu(q)}{8m^3}\gamma^a\gamma^b\gamma^c \right] \\
&\quad \times \left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4} \right) \Bigg] - \delta_2 \frac{1}{2^{\lfloor d_\gamma/2 \rfloor}} \operatorname{tr}\left[\frac{\gamma^a\gamma^a}{2m} \right].
\end{aligned} \tag{110}$$

The counter terms can now easily be extracted from the renormalization condition,

$$\frac{(d-1)(d+2)}{2}\delta_2 = e^2 \int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\left(\frac{1}{q_0^2 + \frac{\vec{q}^4}{4m^2}} \right.$$

$$\times\left(-\frac{(d-1)(d+2)}{2}\frac{v^2q^2}{q_0^2+v^2q^4}-\frac{(d+2)(d+3)(d-2)}{4}\frac{q^2}{4m^2}\frac{1}{q_0^2+v^2q^4}\right)$$
$$-\frac{1}{\left(q_0^2+\frac{\vec{q}^4}{4m^2}\right)^2}\left(-\frac{1}{2m^2}\frac{vq^6}{q_0^2+v^2q^4}+\frac{2-d}{4m^4}\frac{q^6}{q_0^2+v^2q^4}\right)\Bigg)$$
$$\delta_2 = 8\pi^2\alpha f_2(1/\sqrt{\rho})\mu^{4-d}\int\frac{d^dq}{(2\pi)^d}\frac{1}{q^4}$$
$$\delta_2 = 8\pi^2\alpha f_2(1/\sqrt{\rho})\left(\frac{1}{8\pi^2\epsilon}+\frac{1}{16\pi^2}\left(\log\frac{4\pi\mu^2}{r^2}-\gamma\right)+O(\epsilon)\right), \tag{111}$$

where,

$$f_2(x) = -\frac{1}{4(d-1)(d+2)(2+x)^2}\Big[\left(d^3+3d^2-20d+20\right)x^3+2\left(d^3+3d^2-12d+4\right)x^2$$
$$+8(d+3)(d-2)x+16\left(d^2+d-4\right)\Big]. \tag{112}$$

So we now have in our hands both of the counter terms pertaining to the fermionic sector.

C. The photon self-energy

Now we turn to the counter terms of the photon sector. Let us recall that the transverse and longitudinal projectors associated with the photon propagator are defined respectively as,

$$P^T_{\mu\nu}(k) = \delta_{\mu\nu}-\frac{k_\mu k_\nu}{k^2},\ P^L_{\mu\nu}(k) = \frac{k_\mu k_\nu}{k^2}, \tag{113}$$

and the ones acting on the spatial components alone as,

$$p^T_{ij}(k) = \delta_{ij}-\frac{k_ik_j}{k^2},\ p^L_{ij}(k) = \frac{k_ik_j}{k^2}. \tag{114}$$

With the aid of these projectors the one-loop renormalized photon self-energy can be written down as,

$$\Pi^{\mu\nu,1\mathrm{L}}(k_0,\vec{k}) = -\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\,\mathrm{tr}\left[e\left[i\delta_{\mu0}+\frac{\delta_{\mu i}\not{g}^i(k+2q)}{2m}\right]\right.$$
$$\times\left[\frac{-i(q_0+k_0)+\frac{\not{g}(q+k)}{2m}}{(q_0+k_0)^2+\frac{|\vec{q}+\vec{k}|^4}{4m^2}}\right]$$
$$\times e\left[i\delta_{\nu0}+\frac{\delta_{\nu i}\not{g}^i(k+2q)}{2m}\right]$$
$$\left.\times\left[\frac{-iq_0+\frac{\not{g}(q)}{2m}}{q_0^2+\frac{q^4}{4m^2}}\right]\right]$$

$$
\begin{aligned}
&- \int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\,\mathrm{tr}\left[2e^2\frac{\sqrt{\frac{d}{2(d-1)}}\lambda_a^{ij}\delta_{\mu i}\delta_{\nu j}\gamma^a}{2m}\left[\frac{-iq_0+\frac{g_a(q)\gamma^a}{2m}}{q_0^2+\frac{g_a(q)g_a(q)}{4m^2}}\right]\right]\\
&-\delta_3\left(k^2P^T_{\mu\nu}(k)-k^2\delta_{\mu i}\delta_{\nu j}p^T_{ij}(k)\right)\\
&-\delta_4\delta_{\mu i}\delta_{\nu j}c^2\Lambda^2k^2p^T_{ij}(k)\\
&-\delta_5\delta_{\mu i}\delta_{\nu j}v^2k^4p^T_{ij}(k).
\end{aligned}
\tag{115}
$$

Differentiating twice, we find k^2 in the 00 entry from $\frac{1}{2}F_{0i}F_{0i}$ and $k^2p^T_{ij}$ in the spatial block from $\frac{1}{4}F_{ij}F_{ij}$, and we therefore read off the coefficient of the latter as $p^T_{ij}\Pi^{ij}/(d-1)$. The expression further simplifies if we consider that the integrals of the type,

$$
\begin{aligned}
&\int\frac{d^dq}{(2\pi)^d}f(q^2)g_a(q)\\
=&\int\frac{d^dq}{(2\pi)^d}f(q^2)\left(\sqrt{\frac{d}{2(d-1)}}\lambda_a^{ij}q_iq_j\right),\\
=&\int\frac{d^dq}{(2\pi)^d}f(q^2)\sqrt{\frac{d}{2(d-1)}}\left(\lambda_a^{ij}q^2\delta_{ij}+\cancel{\left(q_iq_j-q^2\delta_{ij}\right)}\right),\\
=&0,
\end{aligned}
\tag{116}
$$

which implies that the contribution from the four-point vertex disappears from the photon self-energy. It is also more convenient to consider the spatial components of the photon self-energy, so that we can easily project out the longitudinal part within the spatial indices.

With the third renormalization condition Eq. (94) in mind we evaluate,

$$
\begin{aligned}
\Pi^{00,1\mathrm{L}}(0,\vec{k})=&-\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\,\mathrm{tr}\left[-e^2\left[\frac{-iq_0+\frac{g_a(q+k)\gamma^a}{2m}}{q_0^2+\frac{g_a(q+k)g_a(q+k)}{4m^2}}\right]\right.\\
&\left.\times\left[\frac{-iq_0+\frac{g_a(q)\gamma^a}{2m}}{q_0^2+\frac{g_a(q)g_a(q)}{4m^2}}\right]\right]-\delta_3k^2,\\
=&2^{\lfloor d_\gamma/2\rfloor}e^2\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\frac{-q_0^2+\frac{d}{d-1}\frac{\left\{(\vec{k}+\vec{q})\cdot\vec{q}\right\}^2-\frac{1}{d}|\vec{q}+\vec{k}|^2q^2}{4m^2}}{\left(q_0^2+\frac{q^4}{4m^2}\right)\left(q_0^2+\frac{|\vec{q}+\vec{k}|^4}{4m^2}\right)}-\delta_3k^2,\\
=&\frac{2^{\lfloor d_\gamma/2\rfloor}de^2m}{d-1}\int\frac{d^dq}{(2\pi)^d}\frac{\left(\vec{q}\cdot\vec{k}\right)^2-q^2k^2}{q^2|\vec{k}+\vec{q}|^2\left(|\vec{k}+\vec{q}|^2+q^2\right)}-\delta_3k^2,
\end{aligned}
\tag{117}
$$

$$
\begin{aligned}
=&-\frac{2^{\lfloor d_\gamma/2\rfloor}de^2m}{d-1}\int\frac{d^dq}{(2\pi)^d}\frac{k^2\left(1-\cos^2\vartheta_{kq}\right)}{q^2+k^2+2qk\cos\vartheta_{kq}}\\
&\times\frac{1}{2q^2+k^2+2qk\cos\vartheta_{kq}}-\delta_3k^2.
\end{aligned}
\tag{118}
$$

By expanding the integral expression in small k we obtain,

$$
\Pi^{00,1\mathrm{L}}(0,k\hat{e}_1)=-\frac{2^{\lfloor d_\gamma/2\rfloor-1}de^2m}{d-1}\int\frac{d^dq}{(2\pi)^d}\frac{1-\cos^2\vartheta_{kq}}{q^4}k^2-\delta_3k^2+O(k^3/q^5),
\tag{119}
$$

$$= -\frac{2^{\lfloor d_\gamma/2\rfloor-1} d e^2 m}{d-1} \int \frac{d^dq}{(2\pi)^d} \frac{1}{q^4} \left\{ k^2 - \frac{\left(\vec{k}\cdot\vec{q}\right)^2}{q^2} \right\} - \delta_3 k^2 + O(k^3/q^5). \tag{120}$$

It is shown in Sec. II C that for spatial momentum integrals the following relation holds:

$$\int \frac{d^dq}{(2\pi)^d} f(q^2) q_i q_j = \frac{\delta_{ij}}{d} \int \frac{d^dq}{(2\pi)^d} f(q^2) q^2. \tag{121}$$

We can resolve the integral involving momentum dot products using this identity to obtain,

$$\Pi^{00,1\mathrm{L}}(0, k\hat{e}_1) = -2^{\lfloor d_\gamma/2\rfloor-1} e^2 m \int \frac{d^dq}{(2\pi)^d} \frac{1}{q^4} k^2 - \delta_3 k^2 + \ldots. \tag{122}$$

For the cancellation of the ultraviolet divergences the gauge field counter term has to be,

$$\begin{aligned} \delta_3 &= -2^{\lfloor d_\gamma/2\rfloor-1} \tilde{e}^2 \tilde{m} \mu^{4-d} \int \frac{d^dq}{(2\pi)^d} \frac{1}{q^4} \\ &= -C_1\, 8\pi^2 \alpha \left(\frac{1}{8\pi^2\epsilon} + \frac{1}{16\pi^2} \left(\log \frac{4\pi\mu^2}{r^2} - \gamma \right) + O(\epsilon) \right), \end{aligned} \tag{123}$$

where

$$C_1 = 2^{\lfloor d_\gamma/2\rfloor-1}. \tag{124}$$

The same result is obtained by directly taking the double k derivative towards $k \to 0$.[4]

For the fourth and the last renormalization condition we consider the spatial components of the gauge field polarization at zero external frequency,

$$\begin{aligned} \Pi^{ll,1\mathrm{L}}(0,\vec{k}) = &- \int \frac{dq_0}{2\pi} \frac{d^dq}{(2\pi)^d} \,\mathrm{tr} \left[e \left[\frac{\sqrt{\frac{d}{2(d-1)}} \lambda_a^{ij} \gamma^a \delta_{li} \left(k_j + 2q_j\right)}{2m} \right] \left[\frac{-iq_0 + \frac{g_a(q+k)\gamma^a}{2m}}{q_0^2 + \frac{g_a(q+k) g_a(q+k)}{4m^2}} \right] \right. \\ &\left. \times e \left[\frac{\sqrt{\frac{d}{2(d-1)}} \lambda_a^{ij} \gamma^a \delta_{li} \left(k_j + 2q_j\right)}{2m} \right] \left[\frac{-iq_0 + \frac{g_a(q)\gamma^a}{2m}}{q_0^2 + \frac{g_a(q) g_a(q)}{4m^2}} \right] \right] - \delta_5 (\delta_{ll} - 1) \left(v^2 k^4\right), \\ = &- e^2 \int \frac{dq_0}{2\pi} \frac{d^dq}{(2\pi)^d} \,\mathrm{tr} \left[\left[\frac{\gamma^a \left(g_a^l(k) + 2g_a^l(q)\right)}{2m} \right] \left[\frac{-iq_0 + \frac{g_a(q+k)\gamma^a}{2m}}{q_0^2 + \frac{g_a(q+k) g_a(q+k)}{4m^2}} \right] \right. \\ &\left. \times \left[\frac{\gamma^a \left(g_a^l(k) + 2g_a^l(q)\right)}{2m} \right] \left[\frac{-iq_0 + \frac{g_a(q)\gamma^a}{2m}}{q_0^2 + \frac{g_a(q) g_a(q)}{4m^2}} \right] \right] - \delta_5 (\delta_{ll} - 1) \left(v^2 g_a(k) g_a(k)\right). \end{aligned}$$

The integral is a function of the external momentum $\vec{k}$. From the renormalization condition we wish to apply the differential operator Δ^2 on this function and we have also established that,

$$\Delta^2 f(g_a(k+q), g_a^i(k+q')) = \sum_a \frac{\partial^2 f}{\partial g_a(k)^2}. \tag{125}$$

---

[4] The spinor trace $2^{\lfloor d_\gamma/2\rfloor}$ is 4 at $d = 3$ and 16 at $d = 4$.

We shall take the requisite partial derivative in two steps. In the first step we find,

$$\partial_{g_c(k)}\Pi^{ll,1\mathrm{L}}(0,\vec{k}) = -\,e^2\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\,\mathrm{tr}\left[\left[\frac{\gamma^a\left(g_a^l(k)+2g_a^l(q)\right)}{2m}\right]\right.$$
$$\times\left[\frac{\frac{\gamma^c}{2m}}{q_0^2+\frac{g_a(q+k)g_a(q+k)}{4m^2}}-\frac{-iq_0+\frac{g_a(q+k)\gamma^a}{2m}}{\left(q_0^2+\frac{g_a(q+k)g_a(q+k)}{4m^2}\right)^2}\frac{g_c(q+k)}{2m^2}\right]$$
$$\left.\times\left[\frac{\gamma^a\left(g_a^l(k)+2g_a^l(q)\right)}{2m}\right]\left[\frac{-iq_0+\frac{g_a(q)\gamma^a}{2m}}{q_0^2+\frac{g_a(q)g_a(q)}{4m^2}}\right]\right]-\delta_5(\delta_{ll}-1)\left(2v^2g_c(k)\right). \quad (126)$$

In the second step we shall also take $\vec{k}\to 0$ after the derivative. We obtain,

$$\sum_c\partial^2_{g_c(k)}\Pi^{ll,1\mathrm{L}}(0,\vec{k})\Big|_{\vec{k}\to 0} = -\,e^2\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\,\mathrm{tr}\left[\left[\frac{\gamma^a g_a^l(q)}{m}\right]\right.$$
$$\times\left[-\frac{\frac{\gamma^c}{2m}}{\left(q_0^2+\frac{g_a(q)g_a(q)}{4m^2}\right)^2}\frac{g_c(q)}{2m^2}-\frac{\frac{\gamma^c}{2m}}{\left(q_0^2+\frac{g_a(q)g_a(q)}{4m^2}\right)^2}\frac{g_c(q)}{2m^2}\right.$$
$$\left.-\frac{-iq_0+\frac{g_a(q)\gamma^a}{2m}}{\left(q_0^2+\frac{g_a(q)g_a(q)}{4m^2}\right)^2}\frac{\delta_{cc}}{2m^2}+2\frac{-iq_0+\frac{g_a(q)\gamma^a}{2m}}{\left(q_0^2+\frac{g_a(q)g_a(q)}{4m^2}\right)^3}\frac{g_c(q)g_c(q)}{4m^4}\right]$$
$$\left.\times\left[\frac{\gamma^a g_a^l(q)}{m}\right]\left[\frac{-iq_0+\frac{g_a(q)\gamma^a}{2m}}{q_0^2+\frac{g_a(q)g_a(q)}{4m^2}}\right]\right]-\delta_5(\delta_{ll}-1)(d-1)(d+2)v^2,$$
$$= -\,e^2\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\,\mathrm{tr}\left[\left[\frac{\gamma^a g_a^l(q)}{m}\right]\right.$$
$$\times\left[-\frac{\left(-iq_0\frac{(d-1)(d+2)}{4m^2}+\frac{d^2+d+2}{2}\frac{g_a(q)\gamma^a}{4m^3}\right)}{\left(q_0^2+\frac{q^4}{4m^2}\right)^2}+\frac{-iq_0+\frac{g_a(q)\gamma^a}{2m}}{\left(q_0^2+\frac{q^4}{4m^2}\right)^3}\frac{q^4}{2m^4}\right]$$
$$\left.\times\left[\frac{\gamma^a g_a^l(q)}{m}\right]\left[\frac{-iq_0+\frac{g_a(q)\gamma^a}{2m}}{q_0^2+\frac{q^4}{4m^2}}\right]\right]-\delta_5(\delta_{ll}-1)(d-1)(d+2)v^2,$$
$$= -\,e^2\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\,\mathrm{tr}\left[\left[\frac{\gamma^a g_a^l(q)}{m}\right]\right.$$
$$\times\left[\frac{iq_0}{2m^2}\left(\frac{(d-1)(d+2)/2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^2}-\frac{q^4/m^2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^3}\right)\right.$$
$$\left.\left.-\frac{g_a(q)\gamma^a}{4m^3}\left(\frac{\left(d^2+d+2\right)/2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^2}-\frac{q^4/m^2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^3}\right)\right]$$

$$\times \left[\frac{\gamma^a g_a^l(q)}{m}\right]\left[\frac{-iq_0+\frac{g_a(q)\gamma^a}{2m}}{q_0^2+\frac{q^4}{4m^2}}\right]\Bigg] - \delta_5(\delta_{ll}-1)(d-1)(d+2)v^2. \quad (127)$$

From the odd-even nature of the integrand we can separate two quadratures with finite magnitude from this expression. The first quadrature is given by,

$$\begin{aligned}
I_1 &= \int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\,\mathrm{tr}\left[\left[\frac{\gamma^a g_a^l(q)}{m}\right]\left[\frac{\gamma^b g_b^l(q)}{m}\right]\right] \\
&\times \frac{q_0^2}{2m^2}\left(\frac{(d-1)(d+2)/2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^2}-\frac{q^4/m^2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^3}\right)\left(\frac{1}{q_0^2+\frac{q^4}{4m^2}}\right), \\
&= \int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}2^{\lfloor d_\gamma/2\rfloor}\times\frac{(d+2)q^2}{2m^2} \\
&\times \frac{q_0^2}{2m^2}\left(\frac{(d-1)(d+2)/2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^2}-\frac{q^4/m^2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^3}\right)\left(\frac{1}{q_0^2+\frac{q^4}{4m^2}}\right), \\
&= 2^{\lfloor d_\gamma/2\rfloor-2}\frac{(d+2)(d+3)(d-2)}{4m}\int\frac{d^dq}{(2\pi)^d}\frac{1}{q^4}. \qquad (128)
\end{aligned}$$

The second quadrature is given by,

$$\begin{aligned}
I_2 &= \int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\,\mathrm{tr}\left[\left[\frac{\gamma^a g_a^l(q)}{m}\right]\right. \\
&\left.\times -\frac{g_a(q)\gamma^a}{4m^3}\left(\frac{\left(d^2+d+2\right)/2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^2}-\frac{q^4/m^2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^3}\right)\left[\frac{\gamma^a g_a^l(q)}{m}\right]\left[\frac{\frac{g_a(q)\gamma^a}{2m}}{q_0^2+\frac{q^4}{4m^2}}\right]\right]. \qquad (129)
\end{aligned}$$

Now we need the expression for the trace of four γ matrices. The expression is easily obtained through the manipulations,

$$\begin{aligned}
\mathrm{tr}\left[\gamma^a\gamma^b\gamma^c\gamma^d\right] &= \mathrm{tr}\left[\gamma^a\gamma^b(2\delta_{cd}-\gamma^d\gamma^c)\right], \\
&= 2\delta_{cd}\,\mathrm{tr}\left[\gamma^a\gamma^b\right]-\mathrm{tr}\left[\gamma^a\gamma^b\gamma^d\gamma^c\right], \\
&= 2\delta_{cd}\,\mathrm{tr}\left[\gamma^a\gamma^b\right]-2\delta_{bd}\,\mathrm{tr}\left[\gamma^a\gamma^c\right]+\mathrm{tr}\left[\gamma^a\gamma^d\gamma^b\gamma^c\right], \\
&= 2\delta_{cd}\,\mathrm{tr}\left[\gamma^a\gamma^b\right]-2\delta_{bd}\,\mathrm{tr}\left[\gamma^a\gamma^c\right]+2\delta_{ad}\,\mathrm{tr}\left[\gamma^b\gamma^c\right]-\mathrm{tr}\left[\gamma^d\gamma^a\gamma^b\gamma^c\right], \\
\implies \mathrm{tr}\left[\gamma^a\gamma^b\gamma^c\gamma^d\right] &= \delta_{cd}\,\mathrm{tr}\left[\gamma^a\gamma^b\right]-\delta_{bd}\,\mathrm{tr}\left[\gamma^a\gamma^c\right]+\delta_{ad}\,\mathrm{tr}\left[\gamma^b\gamma^c\right], \\
&= 2^{\lfloor d_\gamma/2\rfloor}\left(\delta_{ab}\delta_{cd}-\delta_{ac}\delta_{bd}+\delta_{ad}\delta_{bc}\right).
\end{aligned}$$

We separate the trace related numerator and evaluate to find,

$$\begin{aligned}
&g_a^l(q)g_b(q)g_c^l(q)g_d(q)\,\mathrm{tr}\left[\gamma^a\gamma^b\gamma^c\gamma^d\right], \\
=&2^{\lfloor d_\gamma/2\rfloor}\left(g_a^l(q)g_a(q)g_c^l(q)g_c(q)-g_a^l(q)g_b(q)g_a^l(q)g_b(q)+g_a^l(q)g_b(q)g_b^l(q)g_a(q)\right),
\end{aligned}$$

$$=2^{\lfloor d_\gamma/2\rfloor-1}(2-d)q^6, \tag{130}$$

where in the last line we have used the contraction properties of the functions $g_a(q)$ and $g_a^i(q)$ derived above. The second quadrature then simplifies to,

$$\begin{aligned} I_2 &= \int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d} 2^{\lfloor d_\gamma/2\rfloor-2}\frac{(d-2)q^6}{4m^6} \\ &\quad\times\left(\frac{\left(d^2+d+2\right)/2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^2}-\frac{q^4/m^2}{\left(q_0^2+\frac{q^4}{4m^2}\right)^3}\right)\frac{1}{q_0^2+\frac{q^4}{4m^2}}, \\ &= 2^{\lfloor d_\gamma/2\rfloor-2}\frac{\left(3d^2+3d-14\right)(d-2)}{4m}\int\frac{d^dq}{(2\pi)^d}\frac{1}{q^4}. \end{aligned} \tag{131}$$

Matching the renormalization condition we now obtain the fourth counter term to be,

$$\begin{aligned} \delta_5(d-1)^2(d+2)v^2 &= -e^2\left(I_1+I_2\right), \\ &= -2^{\lfloor d_\gamma/2\rfloor-2}\frac{e^2}{4m}\left((d+2)(d+3)(d-2)+\left(3d^2+3d-14\right)(d-2)\right) \\ &\quad\times\int\frac{d^dq}{(2\pi)^d}\frac{1}{q^4}, \\ \implies \delta_5 &= -\frac{e^2m}{v^2m^2}\left(2^{\lfloor d_\gamma/2\rfloor-2}\frac{(d-2)\left(d^2+2d-2\right)}{(d-1)^2(d+2)}\right)\int\frac{d^dq}{(2\pi)^d}\frac{1}{q^4}, \\ &= -8\pi^2\alpha\frac{1}{\rho}C_2\mu^{4-d}\int\frac{d^dq}{(2\pi)^d}\frac{1}{q^4}, \\ &= -C_2\,8\pi^2\alpha\frac{1}{\rho}\left(\frac{1}{8\pi^2\epsilon}+\frac{1}{16\pi^2}\left(\log\frac{4\pi\mu^2}{r^2}-\gamma\right)+O(\epsilon)\right), \end{aligned} \tag{132}$$

where

$$C_2 = 2^{\lfloor d_\gamma/2\rfloor-2}\frac{(d-2)\left(d^2+2d-2\right)}{(d-1)^2(d+2)}. \tag{133}$$

We can once again pick out the divergence piece above within the MS scheme.[5] Together $\delta_1, \delta_2, \delta_3$, and δ_5 are the minimum number of counter terms that make the renormalized theory finite within the critical hypersurface.

IV. RENORMALIZATION GROUP EQUATIONS WITH MINIMAL SUBTRACTION

We regulate the loop integrals dimensionally, with the scale μ in place of the shell.

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[5] At $d=4$ this C_2 is $22\cdot 2^{\lfloor d_\gamma/2\rfloor-2}/27$, while the shell scheme gives $5\cdot 2^{\lfloor d_\gamma/2\rfloor}/27$; the factor 22/20 between them is unresolved.

Within the minimal subtraction scheme the multiplicative renormalization constants of our theory were computed to be,

$$\begin{aligned}
Z_1 &= 1 + \frac{\alpha f_1(1/\sqrt{\rho})}{\epsilon}, \\
Z_2 &= 1 + \frac{\alpha f_2(1/\sqrt{\rho})}{\epsilon}, \\
Z_3 &= 1 - \frac{C_1 \alpha}{\epsilon}, \\
Z_4 &= 1 - \frac{C_3 \alpha}{K\,\epsilon}, \\
Z_5 &= 1 - \frac{C_2 \alpha}{\rho\,\epsilon}.
\end{aligned} \tag{134}$$

C_3 is the coefficient of $k^2 p_{ij}^T$ in the spatial photon self-energy divided by $c^2\Lambda^2$. In the shell scheme

$$C_3 = 2^{\lfloor d_\gamma/2 \rfloor - d - 1} \pi^{2-d/2} \frac{d\left(3d^3 - d^2 - 4d - 4\right)}{(d-1)^2 \Gamma(d/2+2)} \Lambda^{d-4}, \tag{135}$$

which is $52/9$ at $d = 4$; its dimensional-scheme extraction is still to be done. From the structure of the renormalization constants it is apparent that the renormalized theory is controlled by the two dimensionless couplings α and ρ which will flow with the renormalization scale as the counter terms absorb the divergences of the theory order by order. We are primarily interested in the renormalization of these couplings and the various anomalous critical exponents of the theory.

To see how these couplings flow under the renormalization of the theory we have to note that their bare magnitudes are independent of the synthetic renormalization scale. Let us first consider the renormalized coupling $\alpha = \tilde{e}^2\tilde{m}/(8\pi^2)$. In terms of the bare parameters and the renormalization constants we have the relation,

$$\begin{aligned}
\alpha_B &= \frac{e_B^2 m_B}{8\pi^2}, \\
&= \frac{Z_e^2}{Z_{m^{-1}}} \mu^{4-d} \alpha, \\
&= \frac{Z_1}{Z_3 Z_2} \mu^{4-d} \alpha, \\
&= Z_\alpha \mu^{\Delta_\alpha} \alpha,
\end{aligned} \tag{136}$$

with $\Delta_\alpha = \epsilon$ being the tree-level scaling dimension of the coupling. Similarly, for the photon stiffness $\rho = \tilde{m}^2\tilde{v}^2$ we have,

$$\begin{aligned}
\rho_B &= m_B^2 v_B^2, \\
&= \frac{Z_5}{Z_3} \frac{Z_1^2}{Z_2^2} \mu^0 \rho, \\
&= Z_\rho \mu^{\Delta_\rho} \rho,
\end{aligned} \tag{137}$$

with $\Delta_\rho = 0$ being its tree-level scaling dimension. Now we shall consider the $\log \mu$ derivative of the renormalization scale independent quantities $\log \alpha_B$ and $\log \rho_B$. We obtain two coupled equations,

$$0 = \Delta_\alpha + \left(\beta(\alpha)\partial_\alpha + \beta(\rho)\partial_\rho\right) \log Z_\alpha + \frac{1}{\alpha}\beta(\alpha),$$

$$0 = \Delta_\rho + (\beta(\alpha)\partial_\alpha + \beta(\rho)\partial_\rho)\log Z_\rho + \frac{1}{\rho}\beta(\rho), \tag{138}$$

where $\beta(\alpha) = \frac{d\alpha}{d\log\mu}$ and $\beta(\rho) = \frac{d\rho}{d\log\mu}$ are the desired beta functions for the couplings. Furthermore, we have the loop expansion structure,

$$Z_X = 1 + \sum_{n=1}^{\infty} \frac{X^{(n)}}{\epsilon^n}, \quad X \in \{\alpha, \rho\}, \tag{139}$$

where we have determined the one loop contributions. We can formally expand $\log Z_X$ to obtain,

$$\log Z_X = \log\left(1 + \sum_{n=1}^{\infty} \frac{X^{(n)}}{\epsilon^n}\right) = \sum_{n=1}^{\infty} \frac{\delta_X^{(n)}}{\epsilon^n}. \tag{140}$$

Solving the coupled equations order by order in $1/\epsilon$, and demanding that the beta functions stay finite as $\epsilon \to 0$, gives

$$\begin{aligned}\beta(\alpha) &= -\epsilon\alpha + \left(f_1(1/\sqrt{\rho}) - f_2(1/\sqrt{\rho}) + C_1\right)\alpha^2, \\ \beta(\rho) &= 2\left(f_1(1/\sqrt{\rho}) - f_2(1/\sqrt{\rho}) + \frac{C_1}{2}\right)\alpha\rho - C_2\alpha.\end{aligned} \tag{141}$$

Every coefficient above is a function of d alone. The fixed point is the one place where d must be evaluated; at $d = 4$ the Clifford data are $(d-1)(d+2)/2 = 9$ and $2^{\lfloor d_\gamma/2 \rfloor} = 16$, so that $C_1 = 8$ and $C_2 = 88/27$, and we numerically find,[6]

$$\begin{aligned}\alpha^* &= 0.113419\,\epsilon, \\ \rho^* &= 0.338319.\end{aligned} \tag{142}$$

V. SYMMETRY ALLOWED PERTURBATIONS

Cubic symmetry protects the quadratic band touching of the fermions in our model; however, perturbations with rotational anisotropies are symmetry allowed. The couplings to such terms are marginal at the tree level and we wish to compute their anomalous dimensions close to the critical point by considering the operator insertion method. The band anisotropy term can be written as,

$$L_{\text{ba}} = -\lambda s_a \psi^\dagger g_a(\partial, \partial)\gamma^a \psi, \tag{143}$$

where the signs s_a, defined above by the diagonal and off-diagonal Gell-Mann matrices, make the perturbation symmetric under the cubic group [4]. The most generic cubic symmetry allowed U(1) gauge field action[7] is **[ToDo: proper reference here]**,

$$L_A = \frac{1}{2}F_{0i}^2 + \frac{c^2\Lambda^2}{4}F_{ij}F_{ij} + \frac{v^2}{4}F_{ij}\left[-\nabla^2\right]F_{ij} + \frac{v_1}{2}F_{ij}\left[-\frac{\partial^2}{\partial x_i^2}\right]F_{ij} + \frac{v_2}{2}\left(\frac{\partial}{\partial x_i}F_{ij}\right)^2. \tag{144}$$

---

[6] ρ^* is the root of $2\left(f_1(1/\sqrt{\rho}) - f_2(1/\sqrt{\rho}) + C_1/2\right)\rho = C_2$ at $d = 4$, where $f_1 - f_2 = 0.816839$.

[7] We write the action in F_{ij} because $B_i = \frac{1}{2}\epsilon_{ijk}F_{jk}$ is a vector only at $d = 3$, where $B_iB_i = \frac{1}{2}F_{ij}F_{ij}$, $B_i\left[-\nabla^2\right]B_i = \frac{1}{2}F_{ij}\left[-\nabla^2\right]F_{ij}$ and $\sum_i B_i\frac{\partial^2}{\partial x_i^2}B_i = B_i\nabla^2 B_i - \sum_{i,j}F_{ij}\frac{\partial^2}{\partial x_i^2}F_{ij}$.

As discussed before, the second term is a relevant perturbation, and the first and the third term have been considered at the level of the gauge field propagators. The fifth term vanishes identically by the Bianchi identity $\partial_{[i}F_{jk]} = 0$, which at $d = 3$ is the Gauss law for magnetism $\vec{\nabla} \cdot \vec{B} = 0$ and holds everywhere in the absence of any singular magnetic monopoles.**[ToDo: Hunch: at $d = 3$, $\partial_i F_{ij} = \epsilon_{jkl}\partial_k B_l$ is the curl of $\vec{B}$, not its divergence; with $\vec{\nabla} \cdot \vec{B} = 0$ the fifth term equals the v^2 term up to a total derivative rather than vanishing.]** We shall consider the second and the fourth terms here as perturbations to our tree level action.

A. Operator insertions and renormalization

In a renormalized perturbation theory any local operator $O_\alpha(x)$ with engineering dimension $[O_\alpha]$ may mix with other local operators with dimension $\leq [O_\alpha]$ in the process of eliminating divergences [5]. We can consider a perturbing term $L_{\rm pert} = L_{\rm ba} + L_A$, and find the minimal number of necessary counter terms to renormalize the combined action. Alternatively, we can determine whether the perturbing couplings are relevant or irrelevant by calculating the anomalous dimension of the composite operators they couple to. From power counting the engineering dimensions of the composite operators are easily obtained. Operator insertions are understood in terms of renormalized correlation functions,

$$\langle \psi(x_1)\psi^\dagger(x_2)\dots O_\alpha(x)\rangle, \langle A_\mu(x_1)A_\nu(x_2)\dots O_\alpha(x)\rangle,\ \langle O_\alpha(x)O_\alpha(0)\rangle, \tag{145}$$

where the last correlator,

$$\langle O_\alpha(x)O_\alpha(0)\rangle = \frac{1}{|x|^{2\Delta_{O_\alpha}}}, \tag{146}$$

may be taken to define the renormalized scaling dimension of the composite local operator. In our case we are interested in three such perturbing operators,

$$\begin{aligned} O_{\rm ba} &= -\left(s_a \psi^\dagger g_a(\partial,\partial)\gamma^a\psi\right), \\ O_{\rm ga} &= \frac{1}{2}F_{ij}\left(-\frac{\partial^2}{\partial x_i^2}\right)F_{ij}, \\ O_{\rm gm} &= \frac{1}{4}F_{ij}F_{ij}. \end{aligned} \tag{147}$$

In the language of operator insertion, the divergences introduced by a single operator insertion $O_\alpha(x)$ [(145)] are absorbed through multiplicative renormalization of a set of local operators with the same or lower engineering dimensions. The bare and renormalized local operators are related by [5],

$$O_{B,\alpha}(x) = Z_{\alpha\beta}O_\beta(x). \tag{148}$$

Now, within our renormalization scheme the counter terms absorb the UV divergent pieces of the loop integrals, therefore $[Z_{\alpha\beta}] \geq 0$. It is therefore clear that the operator O_α only mixes with operators with $[O_\beta] \leq [O_\alpha]$ [16]. In our particular example we have logarithmic divergences which makes the counter terms dimensionless, therefore the set of operators that mix together and form a basis is,

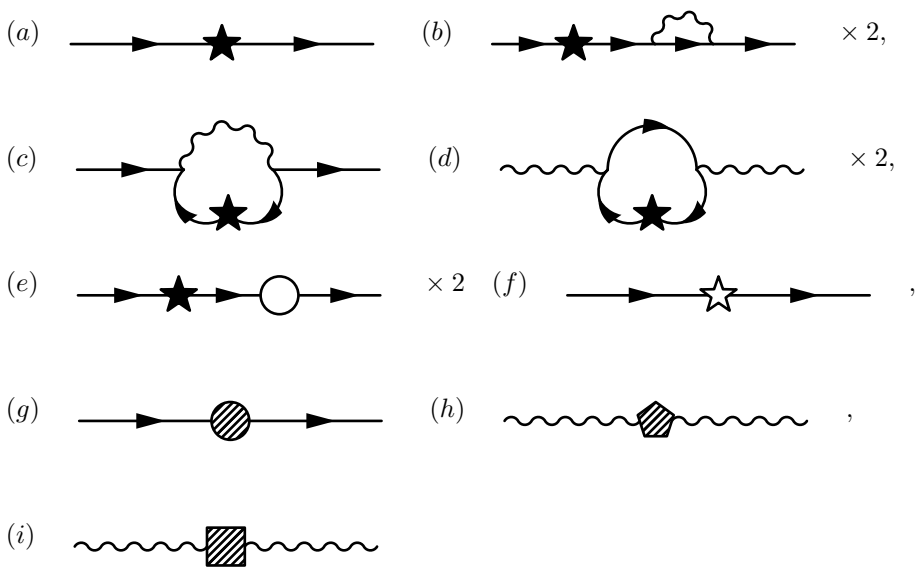
$$O_\alpha \in \left\{ s_a\psi^\dagger g_a\gamma^a\psi, \frac{1}{2}F_{ij}\left(-\frac{\partial^2}{\partial x_i^2}\right)F_{ij}, \psi^\dagger \not{g}\psi, \frac{1}{2}F_{0i}F_{0i}, \frac{1}{4}F_{ij}\left[-\nabla^2\right]F_{ij} \right\}. \tag{149}$$

For the cubic member we make no isotropic subtraction. We multiply its counter term by $\sum_i k_i^4$, whose spherical average is $3k^4/(d+2)$; with the subtraction $\sum_i k_i^4 - 3k^4/(d+2)$ we would only shift v^2, and we defer it to the position-space projection.

B. Insertion of $O_1 = s_a \psi^\dagger g_a \gamma^a \psi$

Diagrammatically, let us denote the operator insertion of O_1 by a filled star and its counter-term insertion as an empty star. Through loop corrections the operator insertion affects both the fermion and the photon sectors, and therefore involves parallel insertion of the counter terms from all operators belonging to the operator basis [Eq. (149)]. We shall consider the same diagrammatic notation for the counter-terms already introduced at the tree level. For the new counter-terms originating from the operator expansion in Eq. (148), we shall shade the corresponding motifs. For the operator $O_2 = \frac{1}{2} F_{ij} \left(-\frac{\partial^2}{\partial x_i^2} \right) F_{ij}$, we shall denote its insertion with a filled pentagon and its counter-term insertion as an empty pentagon. A shaded pentagon shall be used when the counter term stems from the expansion of a different operator insertion.

The insertion of the operator O_1 in the two-particle correlation function [Eq. (145)] at one loop, along with its associated counter terms, is tabulated as follows:



(150)

Among the diagrams, (150) (a) is finite and may be used to set the renormalization condition, i.e., we shall expect the remaining diverging diagrams and the counter terms to cancel each other. The logarithmic divergence of (150) (b) is absorbed by the regular counter terms of the unperturbed theory in (150) (e). With the notation introduced in Eq. (148) and Eq. (149), we can evaluate the remaining terms in the fermionic sector,

$$X_1(k) = (150)\ (c) + (150)\ (f) + (150)\ (g),$$

$$
\begin{aligned}
&= -\int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\Bigg[e\Big[i\delta_{\mu 0}+\frac{2\not{g}^i_a(k)-\not{g}^i_a(q)}{2m}\delta_{\mu i}\Big]\\
&\times\left[\frac{-i(k_0-q_0)+\not{g}(k-q)/(2m)}{(k_0-q_0)^2+(k-q)^4/(4m^2)}\right]\\
&\times[s_a g_a(k-q)\gamma^a]\\
&\times\left[\frac{-i(k_0-q_0)+\not{g}(k-q)/(2m)}{(k_0-q_0)^2+(k-q)^4/(4m^2)}\right]\\
&\times e\Big[i\delta_{\nu 0}+\frac{2\not{g}^i_a(k)-\not{g}^i_a(q)}{2m}\delta_{\nu i}\Big]\\
&\times\left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)\Bigg]-\delta Z_{11}\left[s_a g_a(k)\gamma^a\right]-\delta Z_{13}\left[\not{g}(k)\right].
\end{aligned}
\tag{151}
$$

For the sake of simplicity we may choose the external frequency to be zero so that the expression becomes,

$$
\begin{aligned}
&X_1(0,\vec{k})\\
&= -\int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\Bigg[e\Big[i\delta_{\mu 0}+\frac{2\not{g}^i_a(k)-\not{g}^i_a(q)}{2m}\delta_{\mu i}\Big]\\
&\times\left[\frac{iq_0+\not{g}(k-q)/(2m)}{q_0^2+(k-q)^4/(4m^2)}\right]\\
&\times[s_a g_a(k-q)\gamma^a]\\
&\times\left[\frac{iq_0+\not{g}(k-q)/(2m)}{q_0^2+(k-q)^4/(4m^2)}\right]\\
&\times e\Big[i\delta_{\nu 0}+\frac{2\not{g}^i_a(k)-\not{g}^i_a(q)}{2m}\delta_{\nu i}\Big]\\
&\times\left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)\Bigg]-\delta Z_{11}\left[s_a g_a(k)\gamma^a\right]-\delta Z_{13}\left[\not{g}(k)\right].
\end{aligned}
\tag{152}
$$

Our renormalization condition implies that $X_1(0,\vec{k})$ is a finite quantity with no remaining UV divergences. We shall reformulate the renormalization condition by considering two operators, $\Delta^s_a = s_a\Delta_a$ with Δ_a the operator defined above and demanding that,

$$
\begin{aligned}
\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\left[\not{\Delta}X_1(0,\vec{k})\right]\Big|_{\vec{k}\to 0} &= 0,\\
\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\left[\not{\Delta}^s X_1(0,\vec{k})\right]\Big|_{\vec{k}\to 0} &= 0.
\end{aligned}
\tag{153}
$$

From the first condition we obtain,**[ToDo: Notation: L_b below is the Δ_b of $\not{\Delta}$, and $d^i(q)$, like $d^\mu_b(q)$ in the band mass counter term, is the older name of $g^i(q)$.]**

$$
\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\left[\gamma^b L_b X_1(0,\vec{k})\right] = -\int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\Bigg[\gamma^b L_b
$$

$$
\begin{aligned}
&\times \left(e\Big[i\delta_{\mu 0} + \frac{\gamma^a \left(2g_a^i(k) - g_a^i(q)\right)}{2m}\delta_{\mu i}\Big] \right. \\
&\times \left[\frac{iq_0 + g_a(k-q)/(2m)\gamma^a}{q_0^2 + g_a(k-q)^2/(4m^2)} \right] \\
&\times [s_a g_a(k-q)\gamma^a] \\
&\times \left[\frac{iq_0 + g_a(k-q)/(2m)\gamma^a}{q_0^2 + g_a(k-q)^2/(4m^2)} \right] \\
&\times e\Big[i\delta_{\nu 0} + \frac{\gamma^a \left(2g_a^i(k) - d^i(q)\right)}{2m}\delta_{\nu i}\Big] \\
&\times \left. \left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2 + v^2q^4} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2 + v^2q^4} \right) \right) \Bigg] \\
&- \delta Z_{11} - \frac{(d-1)(d+2)}{2}\delta Z_{13}. \qquad (154)
\end{aligned}
$$

The computation simplifies by taking the limit $\vec{k} \to 0$. We have the quantity of interest,

$$
\partial_{g_b(k)} \left(\left[\frac{iq_0 + g_a(k-q)/(2m)\gamma^a}{q_0^2 + g_a(k-q)^2/(4m^2)} \right] [s_a g_a(k-q)\gamma^a] \left[\frac{iq_0 + g_a(k-q)/(2m)\gamma^a}{q_0^2 + g_a(k-q)^2/(4m^2)} \right] \right) \Bigg|_{\vec{k}\to 0}, \qquad (155)
$$

which can be split into odd and even parts to retain only what gives a finite contribution to the final integral. After some algebra we find that the possibly finite contribution from this piece is given by,

$$
\begin{aligned}
&- q_0^2 \left[\frac{s_b\gamma^b}{\left(q_0^2 + q^4/(4m^2)\right)^2} - \frac{s_a g_a(q)\gamma^a}{\left(q_0^2 + q^4/(4m^2)\right)^3}\frac{g_b(q)}{m^2} \right] \\
&+ \frac{1}{4m^2}\left(\frac{\delta_{ab}g_c(q)g_d(q) + \delta_{bc}g_a(q)g_d(q) + \delta_{bd}g_a(q)g_c(q)}{\left(q_0^2 + q^4/(4m^2)\right)^2} - \frac{g_a(q)g_c(q)g_d(q)}{\left(q_0^2 + q^4/(4m^2)\right)^3}\frac{g_b(q)}{m^2} \right) s_c\gamma^a\gamma^c\gamma^d. \qquad (156)
\end{aligned}
$$

Splitting up the integral into smaller pieces I_3 and I_4, such that,

$$
\frac{1}{2^{\lfloor d_\gamma/2 \rfloor}} \operatorname{tr}\left[\not{\Delta} X_1(0, \vec{k}) \right] \Big|_{\vec{k}\to 0} = 0 = I_3 + I_4 - \delta Z_{11} - \frac{(d-1)(d+2)}{2}\delta Z_{13}, \qquad (157)
$$

we have,

$$
\begin{aligned}
I_3 = &- \int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\frac{1}{2^{\lfloor d_\gamma/2 \rfloor}} \operatorname{tr}\left[\gamma^b \left(e\Big[i\delta_{\mu 0} - \frac{\gamma^a g_a^i(q)\delta_{\mu i}}{2m}\Big] \right. \right. \\
&\times -q_0^2 \left[\frac{s_b\gamma^b}{\left(q_0^2 + q^4/(4m^2)\right)^2} - \frac{s_a g_a(q)\gamma^a}{\left(q_0^2 + q^4/(4m^2)\right)^3}\frac{g_b(q)}{m^2} \right] \\
&\times \left. e\Big[i\delta_{\nu 0} - \frac{\gamma^a g_a^i(q)}{2m}\delta_{\nu i}\Big] \right.
\end{aligned}
$$

$$\times \left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)\Bigg)\Bigg], \tag{158}$$

and,

$$\begin{aligned} I_4 = & -\int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\Bigg[\gamma^b\Bigg(e\Big[i\delta_{\mu 0}-\frac{\gamma^a g_a^i(q)}{2m}\delta_{\mu i}\Big] \\ & \times\frac{1}{4m^2}\left(\frac{\delta_{ab}g_c(q)g_d(q)+\delta_{bc}g_a(q)g_d(q)+\delta_{bd}g_a(q)g_c(q)}{\left(q_0^2+q^4/(4m^2)\right)^2}-\frac{g_a(q)g_c(q)g_d(q)}{\left(q_0^2+q^4/(4m^2)\right)^3}\frac{g_b(q)}{m^2}\right)s_c\gamma^a\gamma^c\gamma^d \\ & \times e\Big[i\delta_{\nu 0}-\frac{\gamma^a g_a^i(q)}{2m}\delta_{\nu i}\Big] \\ & \times\left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)\Bigg)\Bigg]. \end{aligned} \tag{159}$$

To enumerate both of the expressions we shall need the formula for the trace of up to six γ matrices, which can be derived based on the result for the trace of the four γ matrices **[ToDo: show the identity involved]**. Below we use the integral and Gell-Mann identities of Sec. II C.

Using these formulae the integral I_3 is computed to be,

$$I_3 = -e^2 m f_3(1/(mv))\int\frac{d^dq}{(2\pi)^d}\frac{1}{q^4}, \tag{160}$$

and similarly,

$$I_4 = -e^2 m f_4(1/(mv))\int\frac{d^dq}{(2\pi)^d}\frac{1}{q^4}, \tag{161}$$

where $f_3(x)$ and $f_4(x)$ are functions of dimensionless variables. Following the MS scheme, the counter terms are set to cancel out the divergences,

$$\delta Z_{11}+\frac{(d-1)(d+2)}{2}\delta Z_{13} = I_3+I_4 = -\alpha(f_3(1/\sqrt{\rho})+f_4(1/\sqrt{\rho}))\frac{1}{\epsilon}. \tag{162}$$

From the second condition we have,

$$\begin{aligned} \frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\left[\gamma^b L_b^s X_1(0,\vec{k})\right] = & -\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\Bigg[\gamma^b L_b^s \\ & \times\left(e\Big[i\delta_{\mu 0}+\frac{\gamma^a\left(2d_a^\mu(k)-d_a^\mu(q)\right)}{2m}\Big]\right. \\ & \times\left[\frac{iq_0+g_a(k-q)/(2m)\gamma^a}{q_0^2+g_a(k-q)^2/(4m^2)}\right] \\ & \times[s_a g_a(k-q)\gamma^a] \\ & \times\left.\left[\frac{iq_0+g_a(k-q)/(2m)\gamma^a}{q_0^2+g_a(k-q)^2/(4m^2)}\right]\right. \end{aligned}$$

$$\times e\Big[i\delta_{\nu 0}+\frac{\gamma^a\left(2d_a^\nu(k)-d^\nu(q)\right)}{2m}\Big]$$
$$\times\left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)\Bigg]$$
$$-\frac{(d-1)(d+2)}{2}\delta Z_{11}-\delta Z_{13}. \tag{163}$$

Similarly as before the differential operator acts on the part of the integrand that directly depends on the external momenta. In the limit $\vec{k}\to 0$, we evaluate,

$$s_b\partial_{g_b(k)}\left(\left[\frac{iq_0+g_a(k-q)/(2m)\gamma^a}{q_0^2+g_a(k-q)^2/(4m^2)}\right][s_ag_a(k-q)\gamma^a]\left[\frac{iq_0+g_a(k-q)/(2m)\gamma^a}{q_0^2+g_a(k-q)^2/(4m^2)}\right]\right)\Big|_{\vec{k}\to 0}, \tag{164}$$

which can be split into odd and even parts to retain only what gives a finite contribution to the final integral. After some algebra we find that the possibly finite contribution from this piece is given by,

$$-q_0^2\left[\frac{\gamma^b}{\left(q_0^2+q^4/(4m^2)\right)^2}-\frac{s_ag_a(q)\gamma^a}{\left(q_0^2+q^4/(4m^2)\right)^3}\frac{s_bg_b(q)}{m^2}\right]$$
$$+\frac{1}{4m^2}\left(\frac{\delta_{ab}g_c(q)g_d(q)+\delta_{bc}g_a(q)g_d(q)+\delta_{bd}g_a(q)g_c(q)}{\left(q_0^2+q^4/(4m^2)\right)^2}-\frac{g_a(q)g_c(q)g_d(q)}{\left(q_0^2+q^4/(4m^2)\right)^3}\frac{g_b(q)}{m^2}\right)s_bs_c\gamma^a\gamma^c\gamma^d. \tag{165}$$

Just like before, we shall split up the integral into two pieces I_5 and I_6,

$$\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\left[\not{\Delta}^s X_1(0,\vec{k})\right]\Big|_{\vec{k}\to 0}=0=I_5+I_6-\frac{(d-1)(d+2)}{2}\delta Z_{11}-\delta Z_{13}, \tag{166}$$

where we have,

$$I_5=-\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\Bigg[\gamma^b\Bigg(e\Big[i\delta_{\mu 0}-\frac{\gamma^ag_a^i(q)}{2m}\delta_{\mu i}\Big]$$
$$\times -q_0^2\left[\frac{\gamma^b}{\left(q_0^2+q^4/(4m^2)\right)^2}-\frac{s_ag_a(q)\gamma^a}{\left(q_0^2+q^4/(4m^2)\right)^3}\frac{s_bg_b(q)}{m^2}\right]$$
$$\times e\Big[i\delta_{\nu 0}-\frac{\gamma^ag_a^i(q)}{2m}\delta_{\nu i}\Big]$$
$$\times\left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)\Bigg)\Bigg], \tag{167}$$

and,

$$I_6=-\int\frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\frac{1}{2^{\lfloor d_\gamma/2\rfloor}}\,\mathrm{tr}\Bigg[\gamma^b\Bigg(e\Big[i\delta_{\mu 0}-\frac{\gamma^ag_a^i(q)}{2m}\delta_{\mu i}\Big]$$
$$\times\frac{1}{4m^2}\left(\frac{\delta_{ab}g_c(q)g_d(q)+\delta_{bc}g_a(q)g_d(q)+\delta_{bd}g_a(q)g_c(q)}{\left(q_0^2+q^4/(4m^2)\right)^2}-\frac{g_a(q)g_c(q)g_d(q)}{\left(q_0^2+q^4/(4m^2)\right)^3}\frac{g_b(q)}{m^2}\right)$$

$$\times s_b s_c \gamma^a \gamma^c \gamma^d \, e\left[i\delta_{\nu 0} - \frac{\gamma^a g_a^i(q)}{2m}\delta_{\nu i}\right]$$
$$\times \left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2 q^2}{q_0^2 + v^2 q^4} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2 + v^2 q^4}\right)\Bigg)\Bigg]. \tag{168}$$

Following the same steps as before we find the final expression of the counter terms to be,

$$\frac{(d-1)(d+2)}{2}\delta Z_{11} + \delta Z_{13} = -\alpha\left(f_5(1/\sqrt{\rho}) + f_6(1/\sqrt{\rho})\right)\frac{1}{\epsilon}, \tag{169}$$

where $f_5(x)$ and $f_6(x)$ are whole functions of $x = \rho$.[**ToDo: Hunch: they are evaluated at $x = 1/\sqrt{\rho}$, as f_3 and f_4 are.**] From the two renormalization conditions we solve for the individual counter terms and obtain,

$$\delta Z_{11} = \frac{4\left(f_3(1/\sqrt{\rho}) + f_4(1/\sqrt{\rho})\right) - 2(d-1)(d+2)\left(f_5(1/\sqrt{\rho}) + f_6(1/\sqrt{\rho})\right)}{d(d+1)\left(d^2+d-4\right)}\alpha\frac{1}{\epsilon},$$
$$\delta Z_{13} = \frac{4\left(f_5(1/\sqrt{\rho}) + f_6(1/\sqrt{\rho})\right) - 2(d-1)(d+2)\left(f_3(1/\sqrt{\rho}) + f_4(1/\sqrt{\rho})\right)}{d(d+1)\left(d^2+d-4\right)}\alpha\frac{1}{\epsilon}. \tag{170}$$

The insertion of the same operator in the photon sector is captured by (150) (d), (h) and (i). The integral equation describing these graphs is,

$$\begin{aligned}
X_2^{\mu\nu}(k_0,\vec{k}) &= (150)\ (d) + (150)\ (h) + (150)\ (i),\\
&= 2\int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\,\mathrm{tr}\left[e\left[i\delta_{\mu 0} + \frac{\not{g}^i(k) + 2\not{g}^i(q)}{2m}\delta_{\mu i}\right]\right.\\
&\quad\times\left[\frac{-i(q_0+k_0) + \frac{\not{q}(q+k)}{2m}}{(q_0+k_0)^2 + \frac{(q+k)^4}{4m^2}}\right]\\
&\quad\times e\left[i\delta_{\nu 0} + \frac{\not{g}^i(k) + 2\not{g}^i(q)}{2m}\delta_{\nu i}\right]\\
&\quad\left.\times\left[\frac{-iq_0 + \frac{\not{q}(q)}{2m}}{q_0^2 + \frac{q^4}{4m^2}}\right]\times\left[s_a g_a(q)\gamma^a\right]\times\left[\frac{-iq_0 + \frac{\not{q}(q)}{2m}}{q_0^2 + \frac{q^4}{4m^2}}\right]\right]\\
&\quad - \delta Z_{12}\delta_{\mu i}\delta_{\nu j}\left(\epsilon_{kli}\epsilon_{kmj}\left(\frac{k^2}{d-1} - k_k^2\right)k_l k_m\right)\\
&\quad - \delta Z_{14}\left(\delta_{\mu\nu}\delta_{\nu 0}k^2 + \delta_{\mu\nu}\delta_{\nu i}k_0^2 - 2\delta_{\mu 0}\delta_{\nu i}k_0 k_i\right)\\
&\quad - \delta Z_{15}\delta_{\mu i}\delta_{\nu j}k^2\left(\delta_{ij}k^2 - k_i k_j\right).
\end{aligned} \tag{171}$$

To find the contribution to the cubic anisotropy we consider the index contracted form of (150) (d) set at zero external frequency. We can compactly write the expression as,

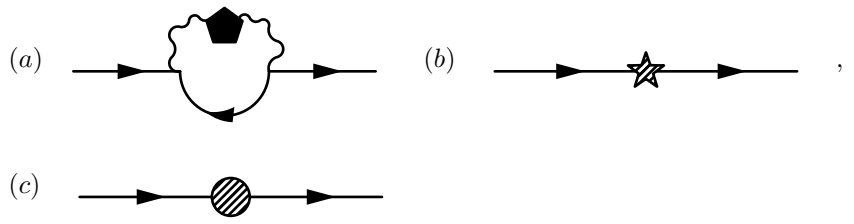
$$X_2^{ii}(0,\vec{k}) = \frac{2e^2}{4m^2}\int\frac{dq_0}{(2\pi)}\frac{d^dq}{(2\pi)^d}\,\mathrm{tr}\left[P(iq_0,\vec{q},\vec{k})M(iq_0,\vec{q})\right] - \delta Z_{12}\sum_i k_i^4 - \delta Z_{15}(d-1)k^4, \tag{172}$$

where,

$$P(iq_0,\vec{q},\vec{k}) = \not{g}^i(k+2q)\left[\frac{1}{-iq_0+\frac{\not{g}(q+k)}{2m}}\right]\not{g}^i(k+2q) \text{ and,}$$
$$M(iq_0,\vec{q}) = \left[\frac{1}{-iq_0+\frac{\not{g}(q)}{2m}}\right][s_a g_a(q)\gamma^a]\left[\frac{1}{-iq_0+\frac{\not{g}(q)}{2m}}\right]. \tag{173}$$

C. Insertion of $O_2 = \frac{1}{2}F_{ij}\left(-\frac{\partial^2}{\partial x_i^2}\right)F_{ij}$

We compute these corrections by considering the following operator inserted diagrams:



(174)

These diagrams evaluate to,

$$X_3(k) = (174)\ (a) + (174)\ (b) + (174)\ (c) \tag{175}$$
$$\begin{aligned}
&= -\int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\left[e\Big[i\delta_{\mu 0}+\frac{2\not{g}^i_a(k)-\not{g}^i_a(q)}{2m}\delta_{\mu i}\Big]\right.\\
&\quad\times\left[\frac{1}{i(k_0-q_0)+\not{g}(k-q)/(2m)}\right]\\
&\quad\times e\Big[i\delta_{\nu 0}+\frac{2\not{g}^i_a(k)-\not{g}^i_a(q)}{2m}\delta_{\nu i}\Big]\\
&\quad\times\left(\delta_{\mu 0}\delta_{\mu' 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\mu i}\delta_{\mu' j}\frac{\delta_{ij}}{q_0^2+v^2q^4}\right)\times\delta_{\mu' i'}\delta_{\nu' j'}\left(\epsilon_{kli'}\epsilon_{kmj'}\left(\frac{q^2}{d-1}-q_k^2\right)q_l q_m\right) \qquad (176)\\
&\quad\times\left.\left(\delta_{\nu' 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2+v^2q^4}+\delta_{\nu' i''}\delta_{\nu j''}\frac{\delta_{i''j''}}{q_0^2+v^2q^4}\right)\right]-\delta Z_{21}\left[s_a g_a(k)\gamma^a\right]-\delta Z_{23}\left[\not{g}(k)\right], \qquad (177)\\
&= -\int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\left[e\Big[\frac{\not{g}^i_a(2k-q)}{2m}\Big]\right.\\
&\quad\times\left[\frac{1}{i(k_0-q_0)+\not{g}(k-q)/(2m)}\right]\\
&\quad\times\left.e\Big[\frac{\not{g}^j_a(2k-q)}{2m}\Big]\right]
\end{aligned}$$

$$\times \left(\frac{1}{q_0^2+v^2q^4}\right) \times \left(\epsilon_{kli}\epsilon_{kmj}\left(\frac{q^2}{d-1}-q_k^2\right)q_l q_m\right)$$
$$\times \left(\frac{1}{q_0^2+v^2q^4}\right)\Bigg] - \delta Z_{21}\left[s_a g_a(k)\gamma^a\right] - \delta Z_{23}\left[\not{g}(k)\right],$$
$$= -\frac{e^2}{4m^2}\int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\Bigg[W(q,k)\left(\frac{1}{q_0^2+v^2q^4}\right) \times \left(\epsilon_{kli}\epsilon_{kmj}\left(\frac{q^2}{d-1}-q_k^2\right)q_l q_m\right)$$
$$\times \left(\frac{1}{q_0^2+v^2q^4}\right)\Bigg] - \delta Z_{21}\left[s_a g_a(k)\gamma^a\right] - \delta Z_{23}\left[\not{g}(k)\right]. \tag{178}$$

[ToDo: $W(q,k)$ in the last line is not defined; from the lines above it is the trace numerator of the two vertices and the fermion propagator.]

D. Anomalous dimension of local operators: Operator insertion

We want to find out the scaling dimension of composite operators such as $O(x) = \psi^\dagger(x)N\psi(x)$, where N are arbitrary matrices from the set $N \in \{1, \gamma^a, \gamma^{ab}\}$. In Euclidean space-time the correlation functions of a local operator O with a well defined scaling dimension Δ_O have the form

$$\langle O(x)O(0)\rangle = \frac{C}{|x|^{2\Delta_O}},$$
$$\implies \langle O(k)O(-k)\rangle = C'|k|^{2\Delta_O - d}. \tag{179}$$

One way to compute the scaling dimension is to calculate this correlation function in a renormalized perturbation theory. We shall instead use our ϵ expansion to determine the multiplicative renormalization of such operators, $O(x) = Z_O\tilde{O}(x)$, such that the renormalized correlators obey the Callan-Symanzik equation,

$$\frac{d\log\langle O(k)O(-k)\rangle}{d\log\mu} = 0,$$
$$\implies \frac{d\log Z_O^2}{d\log\mu} + \frac{1}{\langle\tilde{O}(k)\tilde{O}(-k)\rangle}\left(\frac{\partial}{\partial\log\mu} + \beta(\alpha)\frac{\partial}{\partial\alpha} + \beta(\rho)\frac{\partial}{\partial\rho}\right)\langle\tilde{O}(k)\tilde{O}(-k)\rangle = 0,$$
$$\implies \left(\frac{\partial}{\partial\log\mu} + \beta(\alpha)\frac{\partial}{\partial\alpha} + \beta(\rho)\frac{\partial}{\partial\rho} + 2\eta_O\right)\langle\tilde{O}(k)\tilde{O}(-k)\rangle = 0,$$

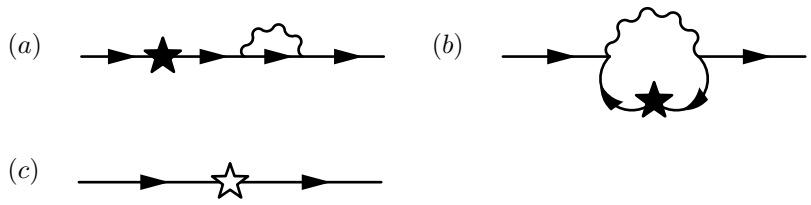
where the anomalous dimension coefficient is given by, $\eta_O = d\log Z_O/d\log\mu$.

We consider the correlation function $\langle\psi(x)\psi^\dagger(x')\left(\psi(y)^\dagger N\psi(y)\right)\rangle$ in a perturbative expansion where the operator $O(y) = \psi^\dagger(y)N\psi(y)$ is inserted only once in all irreducible diagrams. The correlation function includes UV divergences which can be absorbed into the multiplicative renormalization of the renormalized operator $\tilde{O}(y)$ itself and possibly other renormalized local operators of the theory. This is generically expressed in the following manner,

$$O_l(x) = Z_{lm}\tilde{O}_m(x). \tag{180}$$

Here $\{O_l\}$ and $\{\tilde{O}_l\}$ are respectively the unrenormalized and the renormalized local operators of the theory. In this form the renormalized perturbation theory introduces operator mixing.

Now, we specifically consider the insertion of the fermionic bilinear operator $O(x) = \psi^\dagger(x) N \psi(x)$. Diagrammatically, the operator insertion (denoted by a filled star) and the counter-term to cancel the divergences (denoted by an empty star) can be easily compiled.



(181)

The UV divergence of the first diagram is contained within the field strength and coupling renormalization derived above. The second diagram evaluates to,

$$(b) = \int \frac{dq_0}{2\pi} \frac{d^d q}{(2\pi)^d} G(k) C^\mu(k; k-q, q) G(q) G(q) C^\nu(q; q-k, k) D_{\mu\nu}(k-q) G(k). \tag{182}$$

And the counter-term is given by,

$$(c) = G(k) \delta_O(k) G(k), \tag{183}$$

where $\delta_O(k) = \delta Z_{O_l} V^{O_l}(k)$ with a generic mixing of the various gauge invariant local operators $O_l(k) = \psi^\dagger(k) V^{O_l}(k) \psi(k)$. From the structure of the diagrams it is apparent that in this case the local operators of interest are the operator O itself and its various spatio-temporal gradients. Considering the potentially diverging contribution from the diagram (b), we have,

$$G^{-1}(k)(b)G^{-1}(k) \tag{184}$$

$$\begin{aligned} &= \int \frac{dq_0}{2\pi} \frac{d^d q}{(2\pi)^d} \Bigg[e\Big[i\delta_{\mu 0} + \frac{\sqrt{\frac{d}{2(d-1)}} \lambda_a^{ij} \gamma^a \left(k_i + q_i\right) \delta_{\mu j}}{2m} \Big] \\ &\quad \times \left[\frac{-iq_0 + g_a(q)\gamma^a/(2m)}{q_0^2 + q^4/(4m^2)} \right] N \left[\frac{-iq_0 + g_a(q)\gamma^a/(2m)}{q_0^2 + q^4/(4m^2)} \right] \\ &\quad \times e\Big[i\delta_{\nu 0} + \frac{\sqrt{\frac{d}{2(d-1)}} \lambda_a^{ij} \gamma^a \left(k_i + q_i\right) \delta_{\nu j}}{2m} \Big] \\ &\quad \times \left(\delta_{\mu 0}\delta_{\nu 0} \frac{v|\vec{k}-\vec{q}|^2}{(k_0-q_0)^2 + v^2|\vec{k}-\vec{q}|^4} + \delta_{\mu i}\delta_{\nu j} \frac{\delta_{ij}}{(k_0-q_0)^2 + v^2|\vec{k}-\vec{q}|^4} \right) \Bigg]. \end{aligned} \tag{185}$$

[ToDo: Hunch: the temporal photon numerator above, $v|\vec{k}-\vec{q}|^2$, is $v^2|\vec{k}-\vec{q}|^2$, as in the next equation.] It is useful to consider the small k expansion of the above formula. We may generally have,

$$G^{-1}(k)(b)G^{-1}(k) = N K_1\left(1/\epsilon\right) + K_2\left(1/\epsilon\right) \left[ik_0 + \frac{g_a(k)\gamma^a}{2m} \right] + \ldots. \tag{186}$$

The divergences can be absorbed with the counter term,

$$\delta_O(k) = \delta Z_O N + \delta Z'_O \left[ik_0 + \frac{g_a(k)\gamma^a}{2m} \right], \tag{187}$$

or in other words with the operator mixing identity,

$$O(k) = Z_O O(k) + Z_{O'}\psi^\dagger(k)\left[ik_0 + \frac{g_a(k)\gamma^a}{2m}\right]\psi(k). \tag{188}$$

The operator $O'(k) = \psi^\dagger(k)\left[ik_0 + \frac{g_a(k)\gamma^a}{2m}\right]\psi(k)$ has the engineering dimension $\Delta_{O'} = d + z$, whereas $\Delta_O = d$. Therefore, we only need to find Z_O to obtain the anomalous correction to the lowest scaling dimension of the composite operator $O(x)$. So we only need to consider,

$$\begin{aligned} & G^{-1}(0)(b)G^{-1}(0) \qquad (189)\\ =& \int \frac{dq_0}{2\pi}\frac{d^dq}{(2\pi)^d}\left[e\Big[i\delta_{\mu 0} + \frac{\sqrt{\frac{d}{2(d-1)}}\lambda_a^{ij}\gamma^a q_i\delta_{\mu j}}{2m}\Big]\right.\\ & \times\left[\frac{-iq_0 + g_a(q)\gamma^a/(2m)}{q_0^2 + q^4/(4m^2)}\right]N\left[\frac{-iq_0 + g_a(q)\gamma^a/(2m)}{q_0^2 + q^4/(4m^2)}\right]\\ & \times e\Big[i\delta_{\nu 0} + \frac{\sqrt{\frac{d}{2(d-1)}}\lambda_a^{ij}\gamma^a q_i\delta_{\nu j}}{2m}\Big]\\ & \left.\times\left(\delta_{\mu 0}\delta_{\nu 0}\frac{v^2q^2}{q_0^2 + v^2\vec{q}^4} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{q_0^2 + v^2\vec{q}^4}\right)\right]. \qquad (190)\end{aligned}$$

We compute the anomalous dimensions of the local bilinears $O_1 = \psi^\dagger\gamma^a\psi(X)$ and $O_2 = \psi^\dagger\gamma^{ab}\psi(X)$. At the instantaneous photon fixed point, the anomalous dimensions become simpler;

$$\eta_{O_1} = -\frac{2(d^2+d-4)}{d^2+d-2}\alpha^*, \quad \eta_{O_2} = -\frac{8}{d^2+d-2}\alpha^*. \tag{191}$$

We note that for $d = 3$, these anomalous dimensions are twice in magnitude compared to their reported value in Ref. [1].

E. Further perturbations

We consider additional local perturbation to our gauge field action whose anomalous dimension we calculate following the same procedure as above. The perturbing Lagrangian is given by,

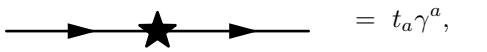
$$\begin{aligned} L_{\text{pert}} =& \frac{1}{2}\left\{A_0\left(\frac{-\partial_0^2 - c^2\Lambda^2\nabla^2 + v^2\nabla^4}{c^2\Lambda^2 - v^2\nabla^2}\right)A_0 + A_i\left(-\partial_0^2 - c^2\Lambda^2\nabla^2 + v^2\nabla^4\right)A_i\right.\\ & \left. - A_0\left(\frac{-\partial_0^2 + v^2\nabla^4}{-v^2\nabla^2}\right)A_0 + A_i\left(-\partial_0^2 + v^2\nabla^4\right)A_i\right\}\\ & + s\left(i\psi^\dagger\psi\right) + t_a\left(\psi^\dagger\gamma^a\psi\right),\\ =& \frac{c^2\Lambda^2}{2}\left\{A_0\left(\frac{\partial_0^2}{v^4\nabla^4}\right)A_0 + A_i\left(-\nabla^2\right)A_i\right\}\end{aligned}$$

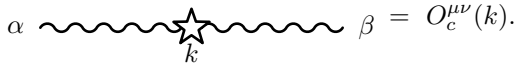
$$+ s\left(i\psi^\dagger\psi\right) + t_a\left(\psi^\dagger\gamma^a\psi\right). \tag{192}$$

The quadratic fermion mass perturbation s is coupled to the conserved current j^0. Therefore, from the conservation law, $\partial_\mu j^\mu = 0$ the dimension of s is fixed in all orders of the perturbation theory to be,

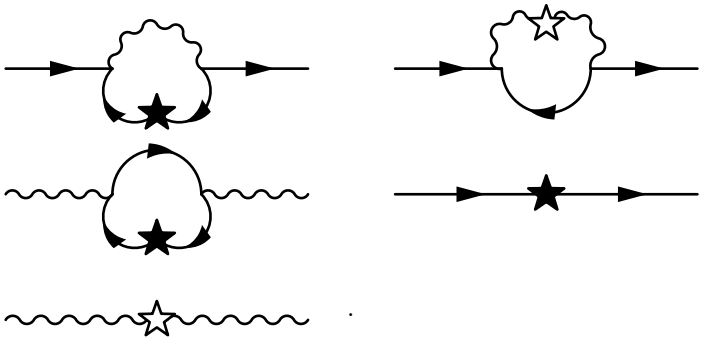
$$[s] = z. \tag{193}$$

The tree-level scaling dimensions $[c^2] = 2z - 2$ and $[t_a] = z$ however receive anomalous correction. Diagrammatically, these are vertices,





We compute these corrections by considering the following operator inserted diagrams:



There are other diagram

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